

Structural Convexity Index

Capturing Flight-to-Quality with an Always-On Long/Short Crisis Strategy

Paul Dimitriadis

Rheon Research

January 2026

This paper introduces the **Structural Convexity Index (SCI)**, a systematic long/short strategy engineered to deliver crisis alpha during material equity drawdowns while maintaining positive long-term drift. Unlike defensive approaches that depend on timing signals or market path—such as trend-following and volatility trading—**SCI is “always-on,”** seeking immediate responsiveness in fast, V-shaped declines by holding permanent durable-minus-fragile spreads. The strategy targets a single, persistent mechanism—**flight-to-quality**—expressed across three uncorrelated sectors through factor-aligned pairs: Gold (long physical gold via GLD; short junior miners via GDXJ), Energy (long integrated majors via XLE; short oilfield services via IEZ), and Credit (long intermediate U.S. Treasuries via IEF; short mortgage REITs via REM). SCI is constructed at 200% long and 100% short (100% net long) and is rebalanced monthly to maintain fixed exposures.

Over the 2007–2025 simulation period, SCI generated a **hypothetical 7.8% CAGR with –84% downside capture** relative to the S&P 500. The strategy simulated positive returns in every major equity drawdown exceeding 15%, including a hypothetical **+24.2% return during the 2022 inflation shock** when stock–bond diversification failed. These results suggest SCI’s convexity is regime-independent, rooted in structural fragility and liquidity-driven repricing rather than falling interest rates. Finally, we argue the strategy’s persistence is supported by a behavioral moat: benchmark-constrained capital struggles to tolerate the tracking error inherent in structural crisis protection.

SCI is a benchmark index for research and measurement purposes only and cannot be invested in directly. Any implementation would require an investment vehicle, the performance of which would differ from the index due to fees, financing/borrow costs, taxes, execution frictions, and operational constraints. Research updates are published at RheonResearch.com.

JEL Classification: G11, G12

Keywords: portfolio construction, tail risk, crisis alpha, long–short equity, structural fragility, convexity, flight to quality, managed futures, asset allocation

EXECUTIVE SUMMARY

The Hedging Dilemma in a New Regime

For decades, investment portfolios leaned on a simple relationship: when equities fall, high-quality bonds rise. The 2022 inflation shock broke that playbook, with stocks and fixed income declining together.

That shift leaves allocators with an uncomfortable trade-off. The most reliable protection—options—tends to be expensive and erodes through premium decay. Lower-cost alternatives—managed futures—can miss fast, V-shaped drawdowns (2018, 2020, 2025) and introduce wide dispersion tied to manager skill, model choices, and execution.

The Systematic Structural Solution

The Structural Convexity Index (SCI) is designed to address this dilemma with a transparent, rules-based long/short framework that removes skill dependency and reduces dispersion risk. The strategy is always-on and maintains 100% net long exposure, seeking to capture flight-to-quality rotations without signals, timing, or discretion—including in rapid, fast-reversing equity events. SCI is designed to deliver:

- **Long-Term CAGR:** Generated by the positive drift of net long exposure to appreciating assets and the structural underperformance of fragile shorts versus durable longs.
- **Crisis Alpha:** Generated by flight-to-quality dynamics—during stress, capital shifts abruptly from fragile business models to durable assets, widening pair spreads.

One Pattern, Three Markets: SCI exploits a single, persistent economic dynamic—**flight-to-quality**—across three uncorrelated sectors:

- **Gold Pair (The Dilution Trade):** Long physical bullion vs. short junior gold miners. This captures the spread between a finite, non-dilutable asset and companies trapped in a "downward dilution spiral" with a high exploration failure rate.
- **Energy Pair (Owners vs. Renters):** Long integrated majors vs. short oilfield services. This pairs the owners of the reserves against service providers who face high operating leverage and capital intensity (the "hamster wheel of Capex").
- **Credit Pair (Stability vs. Leverage):** Long Treasuries vs. short mortgage REITs. This exploits the fragility of entities operating with 6–10x leverage against the safety of government duration.

Crisis-Responsive Returns: Over the 2007–2025 simulation period, this methodology generated a hypothetical **7.8% CAGR** with **–84% downside capture** relative to the S&P 500.

The **strategy simulated positive returns in every major equity drawdown** exceeding 15%, including the **2022 inflation shock (+24.2%)** when traditional stock–bond diversification failed. SCI’s edge relies on business model fragility and flight–to–quality dynamics rather than falling interest rates or specific signals.

The Behavioral Moat: Why the Edge Persists If this structural edge is so visible, why has it not been arbitrated away? The barrier to entry is not informational, but behavioral.

To achieve crisis convexity without paying option premiums or relying on active management, investors pay a different price: **tracking error**. The **strategy captures roughly –18% of the S&P 500’s upside**, often lagging during extended bull markets.

This "behavioral cost" acts as a structural moat. Most professional capital, constrained by quarterly benchmarks and career risk, cannot tolerate multi–year periods of underperformance. Consequently, capital systematically exits the trade during bull markets—precisely before protection is needed most. For allocators capable of viewing the strategy as permanent portfolio infrastructure, this behavioral hurdle preserves the alpha.

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1. Introduction

In the search for "crisis alpha"—positive returns during significant equity market declines—institutional allocators have traditionally turned to active management. Whether through managed futures or volatility trading, the premise remains consistent: skilled managers or sophisticated algorithms can dynamically adjust exposures to profit from market declines.

While these strategies can deliver positive returns during down markets, they share a critical fragility: dependence on informational and execution advantages. Active protection strategies attempt to predict or react to regime shifts faster than the broader market, introducing risks tied to timing, model calibration, and judgment. This dependence on skill and speed not only makes their reliability inherently uncertain but also leads to wide dispersion risk, where performance varies dramatically across managers due to differences in system design, execution, regime adaptation, and even luck—leaving allocators exposed to significant selection errors and inconsistent outcomes. The structure forces allocators to bet on a manager's future decision-making or system design rather than durable economic relationships.

1.1 Engineering a Passive Structural Solution

To deliver consistent crisis alpha without the dispersion inherent in active approaches, a strategy must anchor itself to a pattern; one that emerges in nearly *every* type of severe market decline—not just those that generate the "right" signal for a particular manager or model. Severe equity drawdowns, regardless of their triggers—deflationary panics, funding squeezes, inflation scares, growth shocks, or exogenous events—share one universal commonality: **flight-to-quality**. When liquidity tightens and risk appetite collapses, capital systematically migrates away from leveraged, cash-flow-weak, funding-sensitive exposures toward simpler, stronger, more senior claims. This behavioral and balance-sheet dynamic is not regime-specific; it manifests reliably as investors prioritize durability over fragility, seeking refuge in assets that preserve value amid uncertainty.

The Structural Convexity Index (SCI) is engineered as a **passive, structural solution** to capture this flight-to-quality phenomenon systematically. Unlike active strategies that introduce dispersion through variable skill and timing, SCI maintains an **"always-on" framework**, permanently positioned to respond immediately without waiting for signals or trends to form. This fixed, rules-based structure is transparent, replicable, and free from manager-specific risks—ensuring consistent exposure to the structural edge. At its core, SCI exploits the divergence between durable assets and fragile business models through three carefully constructed long/short pairs across uncorrelated sectors:

- **Gold Pair (The Dilution Trade):** Long physical bullion vs. short junior gold miners—pairing a finite, non-dilutable asset against companies trapped in a "downward dilution spiral" with high exploration failure rates.

- **Energy Pair (Owners vs. Renters):** Long integrated majors vs. short oilfield services—pairing reserve owners against service providers facing high operating leverage and the "hamster wheel of Capex."
- **Credit Pair (Stability vs. Leverage):** Long intermediate Treasuries vs. short mortgage REITs—exploiting the fragility of entities operating with 6–10x leverage against the safety of government duration.

The portfolio is constructed at 200% long / 100% short (100% net long), ensuring positive drift from 100% net long exposure while deriving crisis convexity from the long/short overlay. SCI's rules—set is completely transparent and passive: **fixed-weight pairings with monthly rebalancing**. Because SCI is permanently positioned in these durable–minus–fragile pairings, the crisis engine is armed before the shock occurs. There is no entry signal to miss, no timing gap, no manager skill requirement, and no need for the drawdown to unfold gradually.

1.2 Performance¹

Over the 2007–2025 simulation period, this methodology generated a hypothetical **7.8% CAGR** with **–84% downside capture** relative to the S&P 500. Crucially, it simulated **positive returns in every major equity drawdown exceeding 15%**—including +79% during the GFC, +33% in the COVID–19 crash, and **+24% in the 2022** inflation shock when traditional stock–bond diversification failed.

The remainder of this paper details SCI's construction methodology, examines its historical performance across different crisis regimes and explores why this structural edge has persisted despite its transparency.

2. Methodology

2.1 Conceptual Framework: Capturing Flight–to–Quality

SCI is built around a structural observation: during periods of market stress, capital systematically rotates from fragile business models toward durable assets. This flight–to–quality dynamic manifests as a widening spread between safe haven assets/companies with strong balance sheets and those with operational fragility, high leverage, or funding sensitivity.

The strategy exploits this pattern through permanently held long/short pairs within three distinct sectors. Each pair is constructed to:

1. **Share the same dominant economic factor** (gold prices, energy prices, interest rates) so that idiosyncratic directional views on that factor are minimized.

¹ All performance results included in this paper are hypothetical based on backtested results and do not guarantee future results. See important disclosures in Section 10.

2. **Maximize the durability–fragility contrast** within that factor exposure—pairing the strongest, simplest claims against the most structurally vulnerable business models.
3. **Diversify across uncorrelated crisis channels** so that different types of shocks (deflationary panics, inflation scares, funding squeezes) can activate different pairs within the portfolio.

Because SCI is always positioned in these spreads, the crisis response is immediate—there is no signal to wait for, no trend to detect, and no entry timing required.

2.2 Portfolio Construction

SCI is implemented as a **200% long / 100% short** portfolio, resulting in **100% net long** exposure. This structure serves two functions:

Long–term compounding: The 100% net long exposure provides positive drift from appreciating assets (gold, energy majors, Treasuries). This is the "**baseline return**" of the strategy and is designed to compound over full market cycles.

Crisis convexity: The 100% short exposure is distributed across fragile counterparts within each sector. During stress events, these shorts tend to fall faster and further than their long counterparts, causing the spread to widen. This spread widening is the source of crisis alpha.

The widening of spreads during market stress is not merely a historical observation; it is consistent with the theoretical framework of liquidity pricing established by Vayanos (2004). His model demonstrates that during periods of high volatility, fund managers face tightening performance limits and withdrawals, forcing a 'flight to liquidity.' This structurally increases the risk premium required to hold illiquid or complex assets, causing their prices to collapse relative to liquid collateral. SCI captures this premium by systematically shorting the exact assets (junior miners, mREITs) that suffer most from this liquidity repricing.

The monetary cost of maintaining this structure is primarily **financing and borrow costs** for the long/short overlay, offset by any structural alpha that emerges from the persistent outperformance of durable assets over fragile business models. This cost structure is transparent and largely predictable (although borrowing costs may occasionally spike for certain securities), unlike the path–dependent premiums embedded in option–based hedges.

2.3 The Three Convexity Pairs

Each pair targets a different manifestation of the durability–fragility spread:

Pair 1: Gold—The Dilution Trade

- **Long:** Physical gold (via GLD or similar)
- **Short:** Junior gold miners (via GDXJ or similar)

Rationale: Physical gold is finite and non-dilutable. Junior miners, by contrast, are operationally fragile: they face high exploration failure rates (most deposits never reach production), require continuous equity dilution to fund development, and carry operational leverage that amplifies gold price volatility. During stress, the spread widens as capital rotates from equity–dilution stories toward the underlying metal.

Factor alignment: Both legs respond to gold prices, but the short embeds cash–burn risk, equity dilution, and project execution risk that the long does not.



Figure 1: Cumulative NAV comparison shows GLD appreciating 6–fold while GDXJ remains underwater after 18-years

Pair 2: Energy—Owners vs. Renters

- **Long:** Integrated energy majors (via XLE or similar)
- **Short:** Oilfield services (via IEZ or similar)

Rationale: Integrated majors own reserves—the source of value in energy markets. Oilfield service companies are "renters" who must continuously invest in equipment and chase drilling contracts. They face high operating leverage and capital intensity ("the

hamster wheel of Capex") with no ownership of the underlying resource. When energy markets stress or capital discipline tightens, service providers are structurally disadvantaged.

Factor alignment: Both legs are linked to the energy cycle, but the short is more sensitive to drilling activity, capital spending, and funding conditions.



Figure 2: Cumulative NAV comparison demonstrates XLE appreciation while IEZ experiences persistent structural underperformance

Pair 3: Credit—Stability vs. Leverage

- **Long:** Intermediate U.S. Treasuries (via IEI or similar)
- **Short:** Mortgage REITs (via REM or similar)

Rationale: Treasuries represent the safest duration exposure with zero credit risk. Mortgage REITs operate with 6–10x leverage, hold illiquid assets, and are structurally exposed to spread widening, funding stress, and mark-to-market losses during liquidity shocks. This pair captures the most direct expression of flight-to-quality: when credit stress emerges, capital flees levered credit vehicles for government paper.

Factor alignment: Both legs are interest-rate sensitive, but the short embeds leverage risk, spread risk, and liquidity risk that the long explicitly avoids.

2.4 Implementation: Rules-Based and Transparent

SCI is implemented systematically with no discretionary timing or market forecasting:

Instruments: The strategy uses liquid, exchange-traded funds (ETFs) as proxies for each leg. This ensures transparency, ease of replication, and daily mark-to-market pricing.

Rebalancing: The portfolio is rebalanced monthly back to target weights. This maintains the intended exposure profile and avoids drift in the long/short ratio while minimizing transaction costs and market impact.

Position sizing: Target weights are designed to balance three objectives: (1) diversify the long book across distinct durability themes, (2) provide meaningful crisis responsiveness from each short leg, and (3) stay within defined risk limits (volatility and short-squeeze survivability).

Always-on structure: There is no attempt to tactically enter or exit positions based on valuations, signals, or regime forecasts. The strategy's edge comes from structural positioning, not timing.

Pair	Constituents (Long / Short)	Long Weight	Short Weight	Net Exposure
The Gold Pair	Long Physical Gold (GLD)	60%	-20%	40%
	Short Junior Miners (GDXJ)			
The Credit Pair	Long 3-7yr Treasuries (IEI)	80%	-40%	40%
	Short Mortgage REITs (REM)			
The Energy Pair	Long Energy Majors (XLE)	60%	-40%	20%
	Short Oil Services (IEZ)			
TOTAL		200%	-100%	100%

Note: The portfolio is rebalanced monthly to these target weights to maintain the intended exposure profile.

2.5 Short Weight Methodology

Short weights are set under a risk-first framework intended to maximize the crisis mechanism while limiting day-to-day instability and implementation fragility. The sizing process follows these rules:

1. Spread-first orientation

Shorts are sized so that portfolio risk is not dominated by single-leg volatility. The intent is for outcomes to be primarily driven by *relative* performance within each pair (durable vs fragile), not by noise in one leg.

2. Adverse-move survivability constraint

Short exposure is capped so that a large adverse move in the short leg does not force portfolio deleveraging under extreme stress scenarios.

2.6 Credit Pair Special Case: Asymmetry Override

For the credit pair (IEI / REM), a purely volatility-scaled approach would imply a very small short allocation because REM is materially more volatile than IEI. SCI applies an

explicit **asymmetry override** to this pair because the relevant **stress behavior is not symmetric**:

- In benign environments, REM can rally, but its long-run upside is constrained by a floor in rate spreads.
- In stress environments, REM can decline rapidly and nonlinearly as spreads widen, funding tightens, haircuts rise, and forced deleveraging occurs.

Accordingly, the credit short is maintained at a size intended to preserve the pair's crisis sensitivity to funding-spread shocks, subject to the survivability defined above.

2.7 Why it Works: Two Engines

SCI is designed to produce returns from two components:

Engine 1—Net-long drift: a 100% net long allocation to durable assets provides a baseline return stream over time.

Engine 2—Structural spread: fragile business models can exhibit persistent underperformance versus the durable exposures they depend on (dilution, operating leverage, leverage costs, and capital-cycle dynamics).

$$R_{\text{SCI}} = \underbrace{(0.5 \times R_{\text{Total Longs}})}_{\text{Engine 1: Drift}} + \underbrace{((0.5 \times R_{\text{Total Longs}}) - R_{\text{Total Shorts}})}_{\text{Engine 2: Spread}} - \text{Costs}$$

2.8 The Spark: Crisis Alpha (Flight-to-Quality)

During liquidity shocks, three pressures tend to hit the fragile legs simultaneously: (i) the cost of capital rises, and credit spreads widen, (ii) collateral standards tighten and leverage is reduced, and (iii) capital access deteriorates. At the same time, demand rises for durable assets perceived as safe. Because SCI is already positioned on both sides of this rotation, it responds immediately without relying on any timing signal.

3. SCI Edge: Why it Persists?

SCI's edge is structural rather than informational. It does not depend on forecasting, faster data, or hidden signals. Instead, it exploits persistent divergences between durable assets and fragile business models—differences rooted in balance sheet structure and capital cycle mechanics.

If this effect is systematic and visible in history, why hasn't it been arbitrated away? Three durable barriers prevent this edge from being competed away by large-scale capital. These barriers are not merely empirical observations—they are grounded in established finance theory that explains why structural mispricings can persist even when widely known.

3.1 The Behavioral Moat: Tracking Error as a Barrier to Entry

The primary barrier to entry is behavioral. To harvest crisis convexity without paying option premiums, SCI investors must pay a different price: **tracking error**.

Empirically, SCI captures roughly **-18% of the S&P 500's upside** in rising markets. This creates **long stretches of underperformance** where the strategy looks "boring" or even detractive during multi-year equity bull runs.

For most professional allocators, this is a formidable barrier due to **career risk**. Institutional mandates are typically reviewed quarterly against benchmarks. A strategy that visibly lags during a bull market—no matter how robust its crisis mechanics—creates immense pressure to redeem just before protection is needed.

This "behavioral pain" is the **moat**. It prevents the trade from becoming crowded. If the strategy were easy to hold, the alpha would be arbitrated away. The premium exists precisely because most capital cannot tolerate the discipline required to earn it.

Theoretical foundation: Shleifer and Vishny (1997) formalize this mechanism in their seminal paper on the limits of arbitrage. They demonstrate that professional arbitrageurs managing other people's money face a fundamental constraint: investors evaluate performance over short horizons and withdraw capital when returns lag. This creates "synchronization risk"—arbitrageurs may be forced to liquidate positions precisely when mispricings are largest and fundamental convergence is imminent. As they note, "arbitrage requires capital, and capital tends to become scarce at the very time when arbitrage opportunities become plentiful."

For SCI, this dynamic is particularly acute. The strategy systematically underperforms during extended bull markets—exactly when investor patience is thinnest and pressure to redeem is highest. The arbitrageurs who could eliminate the fragility premium are prevented from doing so by the career risk of multi-year tracking error. This is not a temporary market inefficiency—it is a permanent structural barrier rooted in the principal-agent problem between fund managers and their investors.

3.2 The Counterparty: Retail "Lottery Ticket" Demand

Every short position requires a buyer on the other side. Who keeps buying junior miners and levered mREITs despite their structural underperformance?

Behavioral finance literature documents a persistent 'lottery preference' among retail investors, who systematically overpay for assets with high positive skewness regardless of the negative expected return. This creates a structural supply of overvalued assets to which SCI acts as a counterparty.

- **The Junior Miner Dynamic:** These equities are often marketed on "blue sky" potential. Retail investors systematically overpay for the slim chance of a major discovery, ignoring the extremely high probability of dilution and failure.

- **The Yield Trap:** Similarly, retail investors often chase the headline yield of mREITs, ignoring the leverage risks that manifest during credit cycles.

SCI systematically takes the other side of this non-economic flow. The counterparty is not a rational arbitrageur, but a speculator seeking excitement. As long as human psychology favors lottery-like payouts, this structural supply of "bad longs" is unlikely to vanish.

Empirical validation: Kumar (2009) provides comprehensive empirical evidence for this lottery preference. Analyzing millions of retail brokerage accounts, he documents that individual investors systematically tilt their portfolios toward stocks with lottery-like characteristics—high idiosyncratic volatility, high idiosyncratic skewness, and low prices. These preferences persist across demographic groups and market conditions. Critically, Kumar shows that lottery-demand stocks deliver significantly negative abnormal returns over subsequent periods, yet retail demand remains consistently high.

This behavioral tendency creates permanent demand for precisely the exposures SCI shorts. Junior gold miners offer exactly the lottery-like profile retail investors seek: low nominal share prices, extreme positive skewness (the "major discovery" dream), and dramatic volatility. The fact that these securities systematically underperform is irrelevant to lottery-seeking investors—they are paying for the excitement and possibility, not expected value. SCI harvests this behavioral premium by providing the liquidity retail investors demand.

3.3 Institutional Architecture: Providing Liquidity to Forced Sellers

Large pools of capital are constrained by mandates, benchmarks, and regulation.

- Long-only equity managers must remain fully invested regardless of valuation.
- Fixed income managers cannot easily short credit or lever safe assets.

These constraints create **forced, procyclical flows**. In benign conditions, capital crowds into yield and complexity (the very exposures SCI shorts). When stress appears, credit spreads widen and leverage limits are hit, forcing institutions to sell at the worst possible time.

SCI is built to provide liquidity to these forced flows. It holds the assets that institutions rush toward (Gold, Treasuries) and shorts the assets they are forced to dump (levered credit, cyclical equity). The strategy profits from the **institutional rigidity** that prevents large players from pivoting until it is too late.

Liquidity spiral mechanism: Brunnermeier and Pedersen (2009) model precisely this dynamic in their analysis of market liquidity and funding liquidity. They show that these two forms of liquidity are mutually reinforcing: when funding conditions tighten (due to rising risk or increased margins), traders must reduce positions, which depresses market liquidity. Lower market liquidity, in turn, increases the risk of holding positions,

causing lenders to tighten funding conditions further. This creates a "liquidity spiral" that amplifies stress.

Critically, this mechanism hits fragile, leveraged positions asymmetrically. As Brunnermeier and Pedersen demonstrate, "funding liquidity can dry up precisely when market liquidity dries up, which is also when most arbitrageurs will need liquidity to meet margin calls or redemptions." Mortgage REITs operating with 6-10x leverage are archetypal victims of this spiral—as credit spreads widen, their mark-to-market losses trigger margin calls, forcing asset sales that further depress prices. Junior miners face similar dynamics as equity financing evaporates during risk-off episodes, forcing dilutive capital raises or project abandonment.

SCI is structurally positioned on the other side of these forced flows. The strategy holds the safe, liquid assets (Treasuries, physical gold) that institutions desperately seek during liquidity spirals, while shorting the fragile, illiquid positions (mREITs, junior miners) that must be dumped. The profits come not from predicting when spirals will occur, but from being permanently positioned to provide liquidity when they inevitably do.

3.4 Why These Barriers Persist

The three persistence mechanisms are not independent—they reinforce each other:

Behavioral constraints (career risk, short-termism) prevent sophisticated capital from arbitraging away the premium, even when the opportunity is visible. **Retail lottery demand** ensures continuous overvaluation of fragile positions, creating persistent alpha opportunities. **Institutional rigidity** (mandates, regulations) generates forced, predictable flows during stress that the strategy can harvest.

Importantly, these are not market "inefficiencies" that will be corrected through learning or competition. They are structural features of financial markets:

- Principal-agent problems (Shleifer and Vishny 1997) are inherent to delegated asset management
- Lottery preferences (Kumar 2009) are rooted in prospect theory and appear stable across cultures and time periods
- Liquidity spirals (Brunnermeier and Pedersen 2009) are mathematical consequences of margin requirements and mark-to-market accounting

For SCI's edge to disappear would require fundamental changes in human psychology, regulatory architecture, and financial market structure—changes that appear neither imminent nor likely. The fragility premium is not a temporary arbitrage opportunity; it is a permanent feature of market ecology.

4. Performance & Crisis Behavior²

The historical simulation from 2007–2025 spans an unusually demanding mix of regimes: the deflationary shock of the GFC, the zero–rate QE era, the biological shock of COVID–19, and the inflationary tightening of 2022.

Over this period, SCI generated a hypothetical **7.8% annualized return**. However, the path of these returns is distinct: the majority of compounding occurs during volatility shocks rather than steady bull market appreciation.

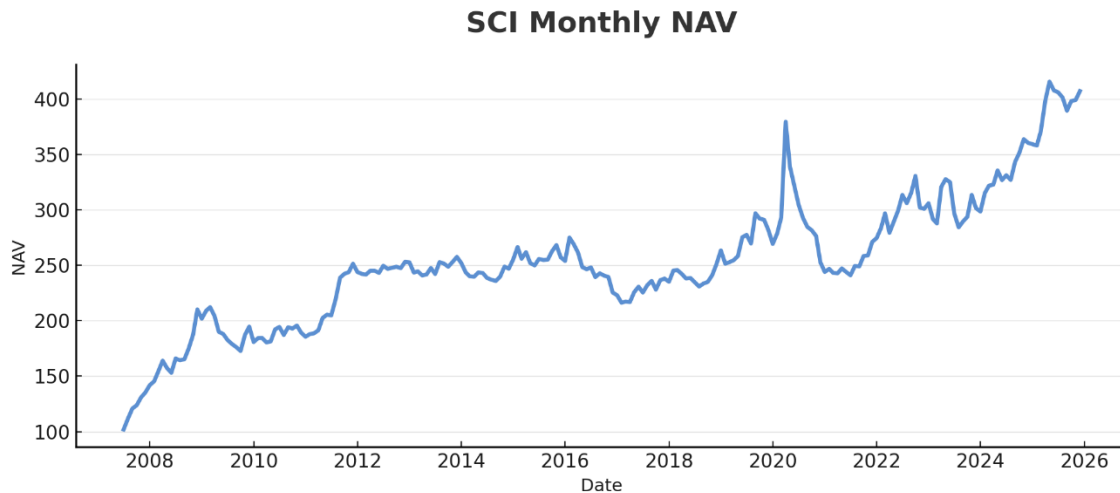


Figure 3: SCI monthly NAV chart. Most of SCI's compounding is concentrated in crisis episodes rather than bull markets.

² All performance results included in this paper are hypothetical based on backtested results and do not guarantee future results. See important disclosures in Section 10.

4.1 Crisis Performance: The Strategy's Core Purpose

A crisis-responsive strategy must be evaluated on its behavior during severe stress, not its Sharpe ratio in quiet markets. SCI methodology simulated **positive returns in every major equity dislocation since 2007**:

Crisis Period	SCI Return	SPY Return	AGG Return	60/40 Return
GFC (Oct 2007-Mar 2009)	+74.9%	-55.5%	+7.4%	-30.2%
COVID (Feb-Mar 2020)	+33.2%	-33.7%	-1.3%	-20.8%
2022 Tightening	+24.2%	-24.5%	-14.4%	-20.5%
Q4 2018	+11.7%	-19.4%	+1.4%	-11.1%
2025 Tariff Correction	+12.6%	-18.8%	+0.9%	-10.9%

Table 1: Crisis performance demonstrating positive SCI returns during five major equity drawdowns exceeding 15% (period measured was SPY peak to trough)

These results span dramatically different crisis types—deflationary collapse (2008), pandemic shock (2020), inflation/tightening (2022), technical correction (2018), and tariff-driven selloff (2025)—confirming the strategy's regime independence.

4.2 The 2022 Test: Validating Structural Edge Over Macro Dependence

The 2022 inflation shock serves as the definitive out-of-sample validation of the strategy's mechanics.

During this period, traditional diversification failed catastrophically. Treasuries fell alongside equities, and the classic 60/40 portfolio suffered its worst year in decades. A strategy relying solely on "flight-to-quality" into bonds would have failed.

SCI, however, returned **+24.2%**. This gain was driven not by duration exposure (which faced headwinds as rates rose) but by the Energy Pair and the Credit Pair.

- **Energy Pair:** Captured the divergence between integrated majors (benefiting from rising commodity prices) and service companies (suffering from cost inflation).
- **Credit Pair:** Profited from widening spreads as mREITs repriced, even as the long Treasury leg declined.

This demonstrates that the strategy's convexity is **not a proxy for falling interest rates**. It works when bonds fail.

4.3 Drawdown Displacement: Swapping Fatal Errors for Tolerable Ones

SCI's most distinctive feature is the **temporal decoupling** of its drawdowns.

Analysis reveals a striking pattern: **none of SCI's five largest drawdowns overlap with major equity crisis windows**. Instead, SCI's drawdowns occur primarily during post-crisis recovery periods (e.g., April 2020 to June 2021) when SPY is rallying strongly.

This inversion is economically critical. It shifts losses from **terminal errors** (compounding losses during a crash) to **tolerable errors** (lagging during a bull market). Losing money when your equity portfolio is up 50% is frustrating but manageable; losing money when your equity portfolio is down 50% is catastrophic.

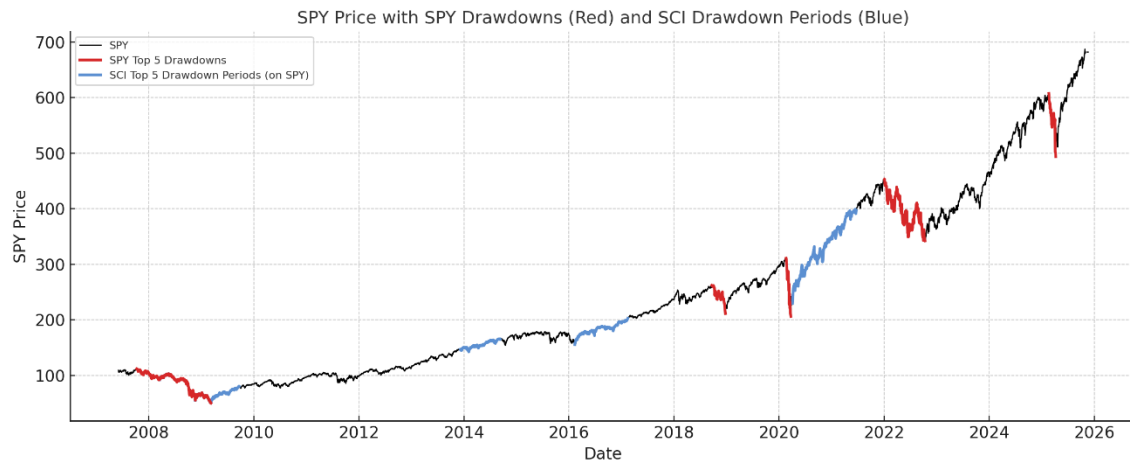


Figure 4: Inverted timing chart showing SPY drawdown periods in red and SCI drawdown periods in blue, demonstrating no overlap

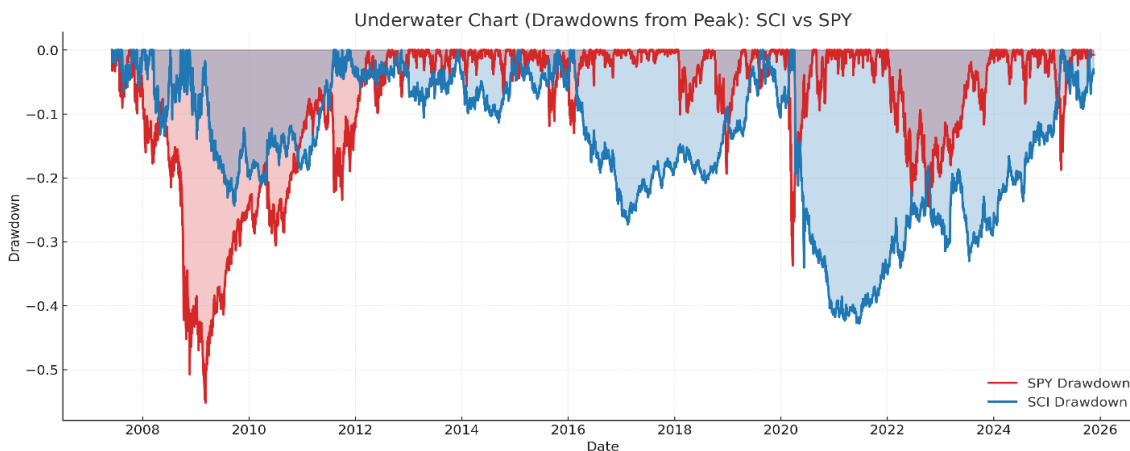


Figure 5: Underwater chart comparing SPY and SCI drawdown profiles demonstrates SCI's inverse performance pattern during major equity declines.

4.4 Correlation Dynamics

Average correlation is a misleading metric for crisis hedges. A strategy that has zero correlation because it oscillates randomly provides no defensive value.

SCI exhibits **State-Dependent Correlation**:

- **Calm Periods:** Correlation to SPY is near zero. The strategy drifts independently.
- **Stress Periods:** When SPY falls >10%, correlation turns sharply negative (−0.4 to −0.65).

This profile—neutral in expansions, negative in stress—contrasts with traditional diversifiers that often see correlations converge to +1.0 during liquidity events.

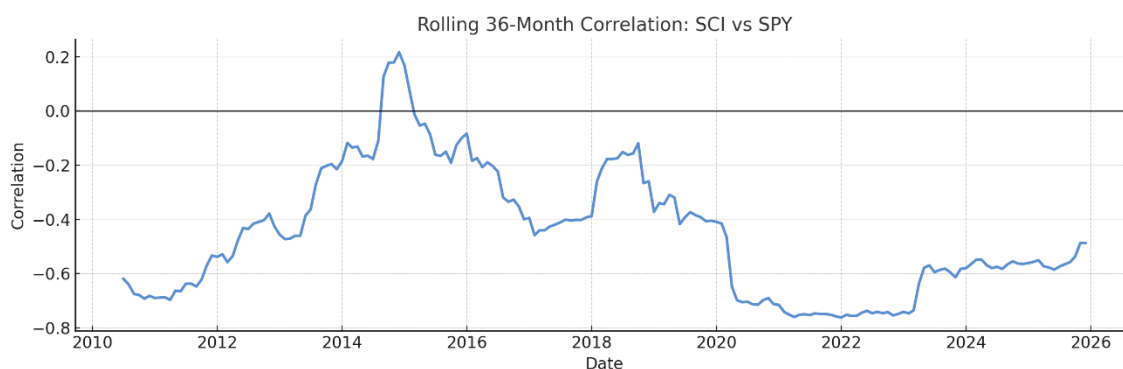


Figure 6: Rolling 36-month correlation between SCI and SPY shows consistent low correlation with frequent negative readings during equity stress periods.

4.5 The Behavioral Premium

Safety is never free. The cost of this convexity is **underperformance during extended bull markets**.

The strategy captures approximately **−18% of the S&P 500's upside**. This creates inevitable tracking error. The implementer's challenge is to tolerate this gap during periods of euphoria. The investor who capitulates during a bull market harvests years of opportunity cost without capturing the crisis gains that justify the strategy's existence.

5. Robustness

Anticipating the Curve-Fitting Objection

A legitimate concern with any systematic strategy is whether historical results depend on fortuitous timing, specific parameter choices, or regime selection. The appearance of "perfect crisis timing" naturally invites skepticism—perhaps the strategy simply got lucky, or perhaps parameters were optimized to historical data in ways that won't persist.

To address this, the strategy was subjected to four independent stress tests: **Monte Carlo reshuffling** (testing luck), **Parameter Sensitivity** (testing optimization), and **Cross-Regime Analysis** (testing adaptability) and **Framework Configurability** (testing structural logic).

5.1 Monte Carlo Reshuffling: Testing Path Dependence

The Question: Did the strategy only work because crises occurred in a specific sequence?

If the strategy's performance relies on the exact timing of past events, it is fragile. To test this, we generated 1,000 synthetic return paths by reshuffling daily returns using bootstrap methods to destroy the historical sequence while preserving the statistical properties of the return distribution.

The Verdict:

- **Performance is Structural:** The realized 7.8% CAGR falls squarely between the 40th and 60th percentiles of the simulated distributions. This indicates that the long-term growth is a property of the return distribution itself, not a lucky sequence of events.
- **Crisis Profile Persists:** The key features—positive skew and right-tail payoffs during stress—persisted across the reshuffled paths. The strategy would likely have exhibited similar defensive characteristics regardless of when the crises occurred.

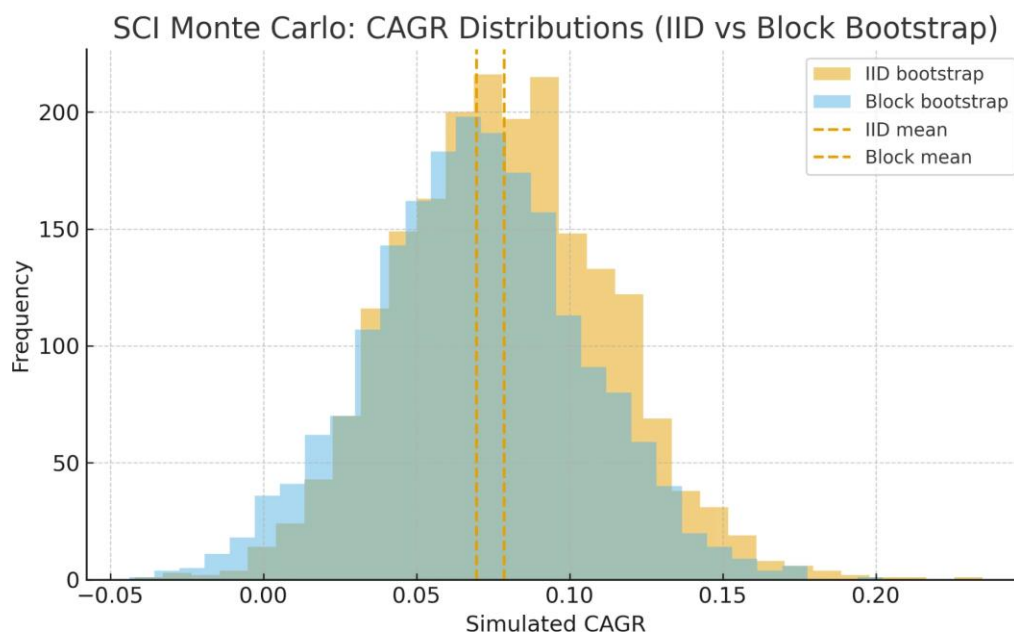


Figure 7: Monte Carlo simulations using both IID and block bootstrap reshuffling confirm SCI's crisis-responsive characteristics are structural rather than path-dependent.

5.2 Parameter Sensitivity Analysis: Avoiding Overfitting

The Question: Are the specific weights (e.g., 80/40) cherry-picked to maximize returns?

If performance collapses when parameters are tweaked slightly, the strategy is overfitted. To assess this, we tested every pair's short ratio at $\pm 20\%$ variations in 5% increments. This produced **729 distinct portfolio configurations** (9 variants per pair $\times$ 3 pairs).

The Verdict:

- **Universal Robustness:** All 729 parameter variants generated positive returns. Not a single configuration failed.
- **Median, Not Maximum:** The published parameters used in SCI produced results in the **59th percentile** for CAGR and **45th percentile** for Sharpe ratio. This confirms that the parameters were not optimized to cherry-pick the extreme right tail of performance; they represent a balanced, middle-of-the-road configuration. The edge derives from the economic relationship between the assets, not the precise weighting.

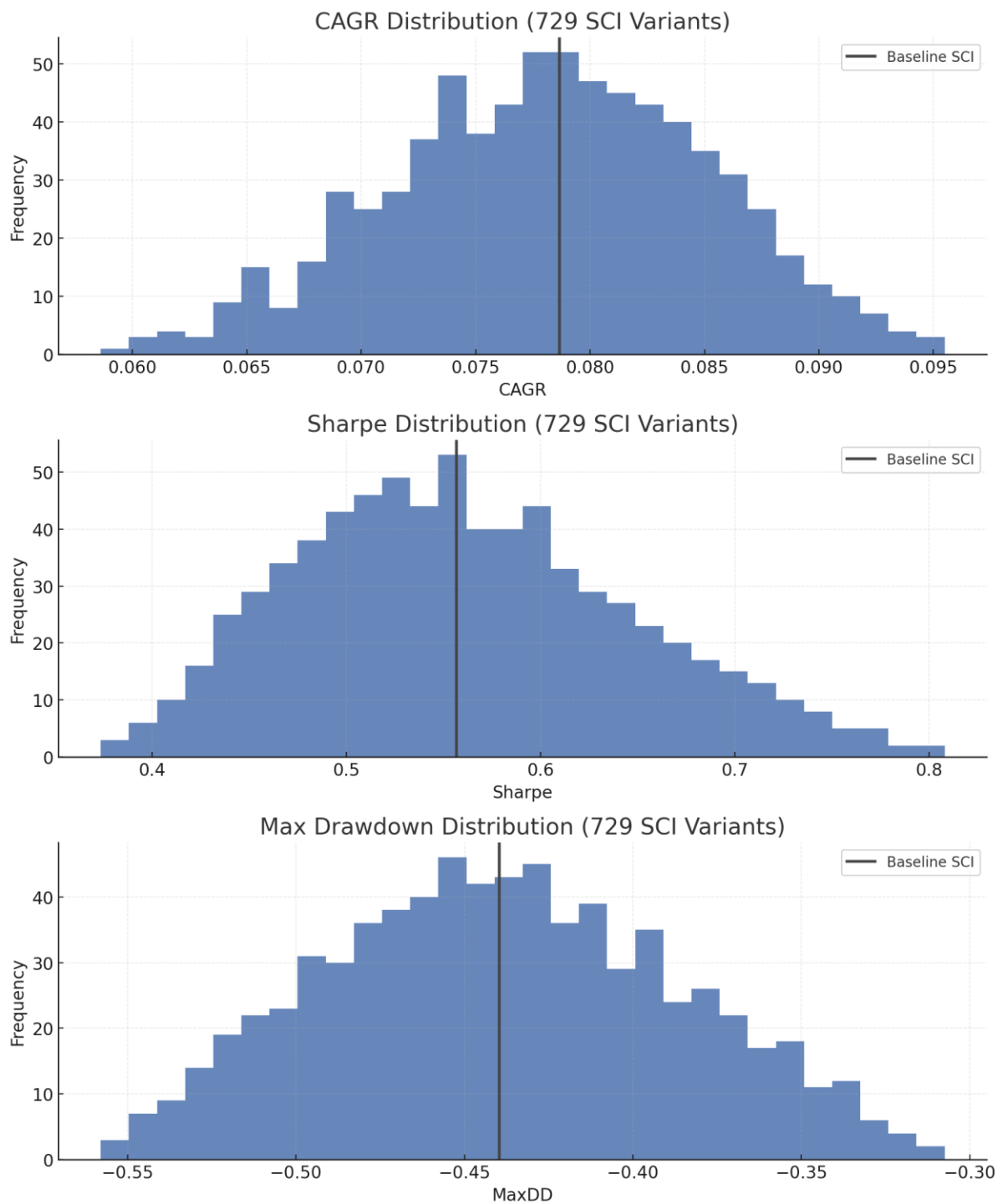


Figure 8: Performance distribution across 729 parameter variants demonstrates robustness, with published parameters producing median outcomes rather than optimized extremes.

5.3 Cross–Regime Stability: Testing Across Different Market Environments

The Question: Does the strategy only work in one type of crisis (e.g., deflation)?

The 2007–2025 period encompasses fundamentally different macroeconomic regimes. A strategy dependent on falling rates (like long–duration Treasuries) or established trends (like trend–following) often fails when the regime shifts.

The Verdict: Analysis confirms consistent crisis responsiveness across four distinct environments:

- **Deflationary Shock (GFC):** Strong protection via the Credit Pair and Gold Pair.
- **QE Expansion:** Modest drift capture during low–volatility appreciation.
- **Pandemic Shock:** Rapid protection via Gold Pair and Credit Pair during a liquidity freeze.
- **Inflationary Tightening (2022):** Strong protection via the Energy Pair and Credit Pair when traditional bonds failed.

This validates that the strategy acts as an "All–Weather" hedge. Because it employs **three distinct transmission mechanisms** (Credit Spreads, Energy Divergence, Safe Haven Flows), it can adapt to crises regardless of whether they are driven by deflation, inflation, or exogenous shocks.

5.4 Framework Configurability: Linearity as Evidence of Structural Logic

The final test distinguishes genuine structural understanding from historical curve–fitting: parameter smoothness. If SCI were optimized to backtested data, we would expect one 'lucky' parameter set with neighbors performing materially worse—creating discontinuities and cliffs as weights change. Instead, we find the opposite.

The Test: We examine three configurations, systematically reducing short exposure: Config 1 (standard weights), Config 2 (80% of short positions), Config 3 (60% of short positions). If the framework reflects genuine structural dynamics, parameter changes should produce proportional, predictable effects.

The Results: Performance changes exhibit near–perfect linearity across all dimensions ($R^2 > 0.99$). Crisis protection degrades smoothly—approximately 8 percentage points per 20% short reduction—while base metrics improve proportionally. CAGR rises from

7.8% to 9.6% to 10.9%; Sharpe ratios from 0.53 to 0.76 to 0.91; correlation to SPY shifts from -0.49 to -0.30 to +0.03. No discontinuities, no regime breaks, no parameter cliffs.

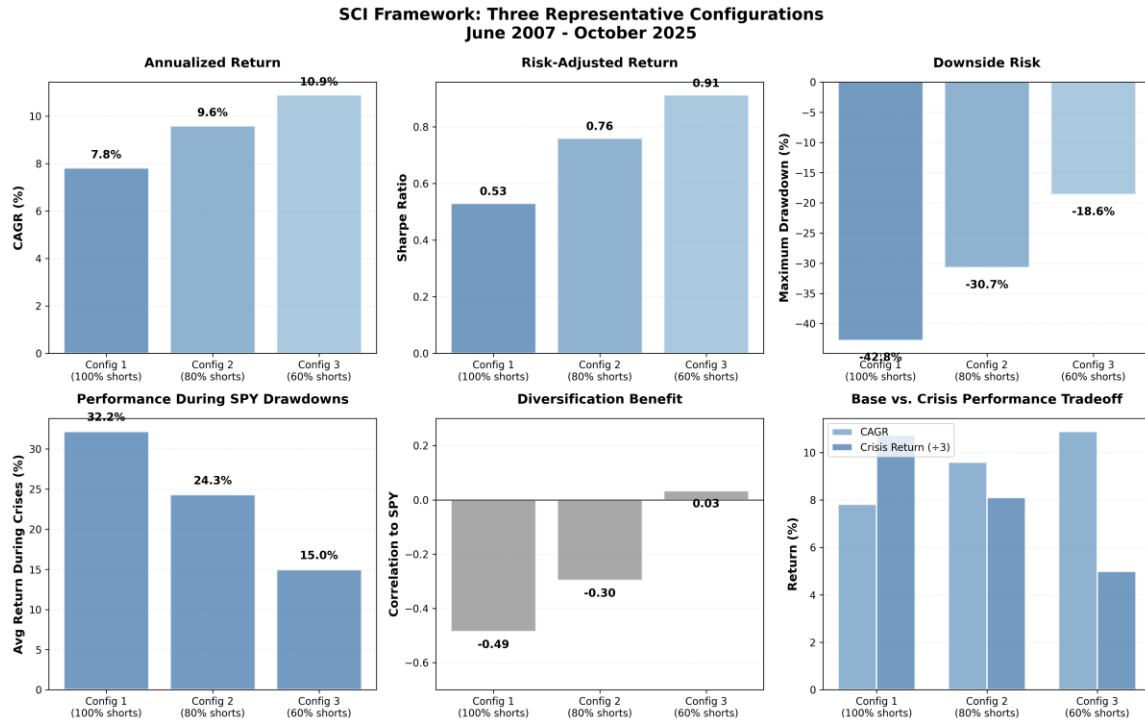


Figure 9: Systematic reduction in short exposure produces proportional, predictable changes across all performance dimensions.

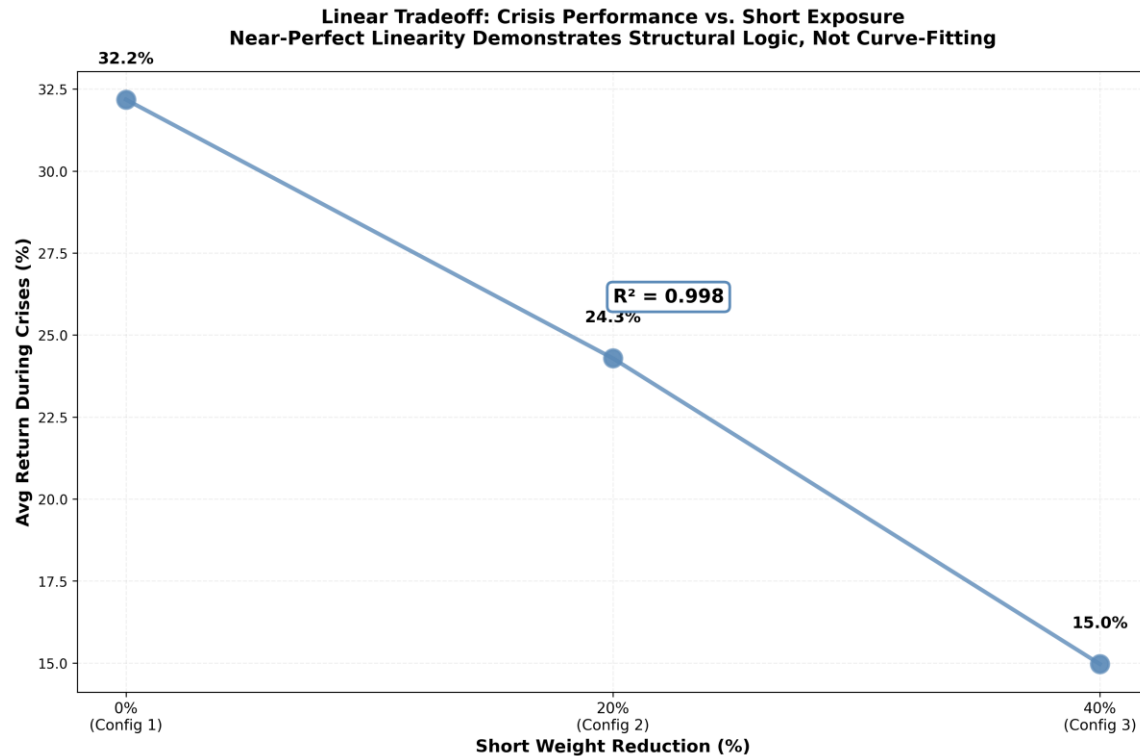


Figure 10. Crisis Performance Degrades Linearly with Short Exposure Reduction ($R^2 = 0.998$), Demonstrating Structural Logic Rather Than Parameter Optimization

The Verdict: Engineering, Not Optimization

This linearity is the signature of genuine structural dynamics. When you manipulate a real economic parameter (short exposure to fragile businesses), outcomes change predictably. When you stumble on a historical artifact, adjacent parameter sets fail unpredictably. The $R^2 = 0.998$ relationship between short reduction and crisis performance demonstrates the former.

Critically, this configurability addresses a practical implementation barrier. Config 1 delivers maximum diversification benefit (-0.49 correlation), but requires tolerating substantial tracking error during bull markets. Many allocators—despite understanding the portfolio math—cannot withstand this career risk. Configs 2 and 3 offer alternatives that sacrifice crisis protection for better stand-alone line-item metrics. The framework makes this tradeoff explicit and quantifiable, enabling selection based on organizational tracking error capacity.

This flexibility also reveals why the opportunity persists: Config 1's exceptional diversification benefit exists precisely because most allocators cannot tolerate the required tracking error. The behavioral barrier to entry—not informational advantage—protects returns. Those with appropriate governance structures and time horizons can harvest this premium; those with tighter constraints can select reduced-short configurations while maintaining positive crisis performance.

6. Portfolio Implementation

Two implementation frameworks address distinct allocator needs: **The 60/40 Replacement** (replacing a portion of fixed income/equity to improve the robustness of the 60/40) and **The Growth Overlay** (stacking SCI on top of existing beta).

6.1 Implementation A: Fixing the 60/40

Target Audience: Allocators seeking to restore the defensive characteristics of balanced portfolios after the failure of bonds in 2022.

This approach involves funding an allocation to SCI by reducing exposure to traditional equities and fixed income. The goal is to introduce a third, uncorrelated asset class that protects when bonds fail.

The 70/30 Solution: Analysis compares a standard 60/40 portfolio against a blended portfolio of **70% Traditional (60/40) and 30% SCI**.

- **Risk Reduction:** The inclusion of SCI reduced the maximum drawdown by more than half, falling from **−35%** (pure 60/40) to **−15%** (blended).
- **Return Enhancement:** Counterintuitively, replacing equity exposure with a lower-beta hedge actually *increased* long-term returns. The blended portfolio generated an **8.6% CAGR** versus 8.1% for the standalone 60/40.

Why it works: This is the result of geometric compounding. By truncating the deep left-tail losses (avoiding the −35% drawdown), the portfolio compounds from a higher base during recoveries.

SCI	CAGR	Vol	Sharpe	Max DD	Terminal
0%	8.1%	9.9%	0.84	−34.7%	414
10%	8.3%	8.3%	1.02	−27.5%	432
20%	8.5%	6.9%	1.23	−21.3%	447
30%	8.6%	5.9%	1.43	−15.0%	457

Table 2: Portfolio metrics for various SCI insertion levels into 60/40 base portfolio

6.2 Implementation B: The Growth Overlay

Target Audience: Aggressive allocators seeking to maximize terminal wealth without increasing tail risk.

This approach uses capital efficiency to "stack" SCI on top of a core equity allocation. Rather than selling assets to buy the hedge, the investor maintains 100% equity exposure and adds SCI as an overlay (e.g., 100% SPY + 50% SCI).

The "Free Lunch" of Negative Correlation: Adding a volatile asset (SCI) to a volatile portfolio (SPY) typically increases risk. However, because SCI captures **−84% of**

downside volatility and correlates negatively during crashes, the combination produces a superior risk-adjusted profile.

- **Scenario:** 100% SPY + 50% SCI Overlay
- **Result:** CAGR increased from 10.5% (SPY alone) to **16.0%** (Stacked).
- **Risk:** Maximum Drawdown *decreased* from -55% (SPY alone) to **-40%** (Stacked).

This configuration increased terminal wealth by **143%** while simultaneously reducing peak-to-trough pain by 27%. This validates SCI's utility as "Portable Alpha"—a return stream that can be transported onto any baseline portfolio to improve its Sharpe ratio.

Overlay	CAGR	Vol	Sharpe	Max DD	Terminal
0%	10.5%	19.6%	0.59	−55.2%	625
20%	12.7%	18.4%	0.72	−48.5%	903
40%	14.9%	17.5%	0.85	−42.7%	1,284
50%	16.0%	17.3%	0.91	−40.3%	1,521

Table 3: Portfolio metrics for various SCI overlay levels on 100% SPY base. As with all return figures in this paper, these are gross hypothetical returns and do not include financing costs or other expenses.

6.3 Behavioral Onboarding Analysis

When SCI is in its top quintile of trailing relative performance to SPY, forward outcomes are materially weaker—more than half of observations produced absolute losses over the subsequent year, and approximately 85% underperformed SPY. The period of greatest external enthusiasm coincides with highest probability of regret-driven abandonment.

7. Comparison to Alternative Defensive Strategies

SCI occupies a distinct position in the spectrum of defensive portfolio strategies. Understanding how it differs from traditional alternatives clarifies when it adds value and when other approaches may be more appropriate.

7.1 SCI vs. Managed Futures (CTAs): The "Speed Factor"

A widely used diversifier for equity drawdowns is managed futures (often implemented via CTA programs). Both SCI and managed futures can exhibit low correlation to equities, but they differ in *how* and *when* they tend to respond during a selloff.

- Managed futures / CTAs (signal-conditional, exposure-adaptive):**
 Many CTA programs adjust positioning based on observed market behavior (often trend or momentum, but also other signals). This can be very effective in drawdowns that *unfold over time* and allow the strategy's signals to develop and persist. In faster, more abrupt declines—especially those that reverse quickly—signals may arrive late, change quickly, or conflict across horizons, which can reduce crisis responsiveness and increase the likelihood of whipsaw.
- SCI (structural, always-on):**
 SCI holds a fixed, continuously maintained structure. It does not require a trend to “confirm” or a model to rotate exposures in response to price action. The portfolio is designed to be in position when stress emerges—when liquidity tightens and the spread between durable and fragile exposures widens—so the intended defensive behavior is not dependent on the timing of a signal.

Evidence in recent drawdowns (this study):

Across the five largest SPY peak-to-trough declines since 2007 examined here, managed futures (proxied by BTOP50) generally held up well but delivered pronounced crisis gains primarily in the slower, more persistent declines (notably 2008 and 2022). In the faster drawdowns (2018, 2020, 2025), the same proxy was closer to flat on average over the peak-to-trough window.

Correlation and complementarity:

Over the analysis period, SCI's correlation to the BTOP50 Index is 0.009. This near-zero relationship suggests SCI is not simply duplicating a managed futures allocation and may provide an independent defensive return stream when used alongside CTAs.

Takeaway:

Managed futures can be an excellent diversifier and may deliver strong drawdown protection when market action aligns with its signals. SCI targets a different mechanism—an always-maintained structure intended to respond during risk-off conditions—and is designed to be particularly relevant when drawdowns are fast and path-dependent.

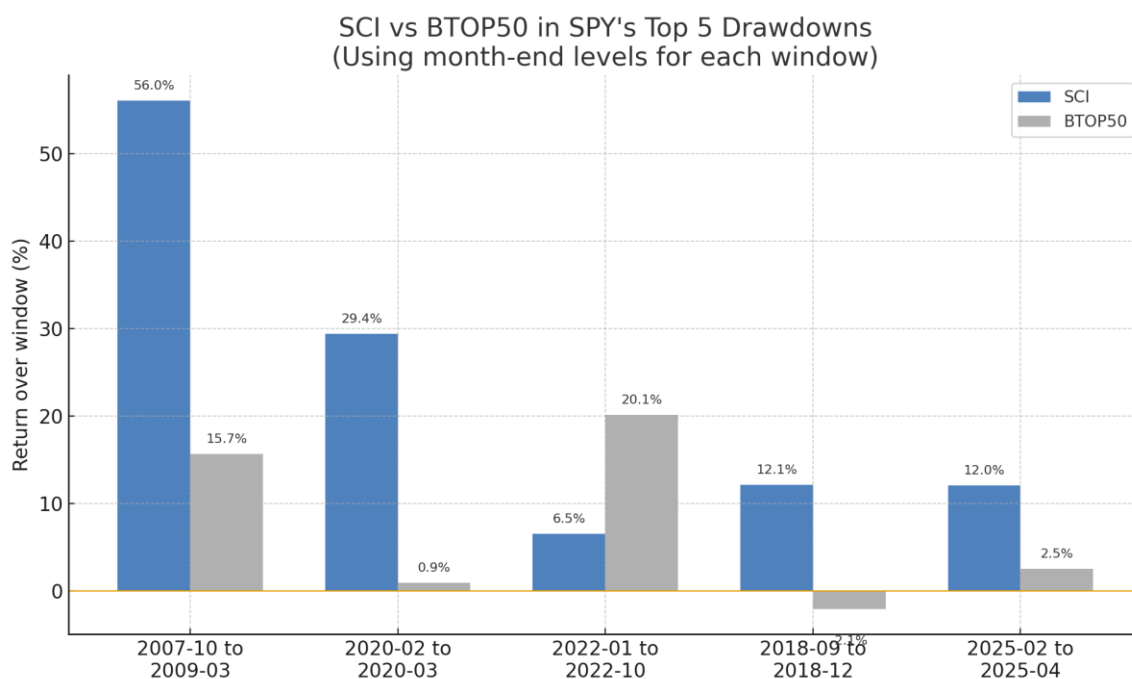


Figure 11: Performance of SCI and BTOP50 across the five largest SPY drawdowns since 2007. Drawdown windows are defined from SPY's daily peak date to its daily trough date; strategy performance is measured over the same windows using month-end series for each strategy.

7.2 SCI vs. Treasury Bonds: The "Correlation Problem"

For decades, long-duration Treasuries were the default hedge for equities. 2022 exposed the fatal flaw in this reliance: interest rate risk.

- **Treasuries** rely on deflation or "flight-to-quality" to generate negative correlation. If a crisis is driven by inflation or rates (like 2022), bonds fail alongside stocks.
- **SCI** relies on **credit spreads** and **asset fragility**. Even if rates are rising (hurting the long Treasury leg), the short positions in mREITs and outperformance of integrated oil producers versus oil service companies can generate enough convexity to offset losses and produce positive returns, as observed in 2022 in this historical sample (+24.2%).

7.3 SCI vs. Option-Based Tail Hedging: The "Cost Problem"

Dedicated tail funds (buying puts) offer the most direct convexity but the highest cost.

- **Option Hedging:** Requires payment of a persistent insurance premium (carry cost of 5–7% annually). This creates a "volatility bleed" that acts as a heavy drag on long-term compounding.
- **SCI:** Funds its protection structurally. By being 100% net long, the strategy aims to be **carry-neutral or carry-positive** over the cycle. The "cost" is paid in tracking error, not cash bleed.

7.4 Summary of Strategies and Differences

Strategy	Primary Mechanism	Best Environment	Primary Weakness
SCI	Structural Fragility	Rapid Shocks & Liquidity Events	Bull Market Lag (Tracking Error)
Managed Futures	Trend Following/Macro	Sustained, Slow-Moving Trends	V-Shaped Reversals (Whipsaw)
Long Treasuries	Duration / Deflation	Deflationary Crises	Inflation / Rising Rate Regimes
Put Options	Convexity	Any Sharp Crash	Permanent Negative Carry (Bleed)

8. Conditions for Failure: When SCI May Not Protect

SCI is not a universal hedge. It is designed to respond to economic crises characterized by funding stress, widening credit spreads, and flight-to-quality flows. This section outlines when that mechanism may not engage, and what would constitute genuine failure rather than expected behavior.

8.1 Failure Mode 1: Market Declines Without Economic Stress

SCI exploits the divergence between durable assets and fragile business models during economic stress. It may provide limited protection in drawdowns driven primarily by technical factors or sector rotation.

Two scenarios illustrate this:

- **Technical disruptions**

Flash crashes, algorithmic breakdowns, or exchange malfunctions that cause abrupt, indiscriminate selling may not trigger the fragility/quality divergence on which SCI relies. The May 2010 “Flash Crash” coincided with European sovereign stress, which happened to support a modest positive SCI outcome; a pure technical malfunction without an economic component might not.

- **Concentration Unwind and Sector Rotation**

A valuation–driven correction in a narrow group of mega–cap leaders could produce a 15–20% SPY drawdown even if the broader economy and credit markets remain healthy. In that case, capital may rotate within equities rather than flee to safety: credit spreads stay tight, safe assets see little bid, and fragile shorts do not experience forced deleveraging.

In such “rotation drawdowns,” SCI is not expected to generate large positive returns. It addresses economic crisis risk in already diversified portfolios, not concentration risk in specific stocks or sectors. The remedy for concentration risk is diversification, not hedging.

8.2 Failure Mode 2: Government Intervention in Short Positions

Direct, targeted government support for SCI’s short exposures could permanently weaken the mechanism. Examples would include:

- Dedicated bailout programs for junior miners
- Ongoing, sector–specific support for oilfield services
- Permanent credit facilities or capital injections targeted at mortgage REITs.

These scenarios appear unlikely. Historically, crisis interventions have focused on systemically important institutions (major banks, dealers, critical infrastructure), not small, fragmented, and non–systemic sectors like junior miners, mid–cap energy services, or mREITs. Notably, these sectors did not receive targeted support even during 2008 or 2020.

By contrast, broad policy responses—market–wide short–selling bans, central bank backstops for credit, or universal liquidity facilities—would likely hurt SCI but represent acceptable “failures.” If authorities succeed in putting a floor under markets and preventing crisis dynamics, the need for crisis protection is reduced by design. In March 2020, for example, SCI’s strong gains during the crash naturally bled back as the Federal Reserve’s interventions truncated the credit crisis; this was the hedge doing its job and then decaying as the fire was extinguished, not a mechanism breakdown.

8.3 Failure Mode 3: Breakdown of Persistence Mechanisms

SCI's edge depends on three persistence mechanisms discussed earlier: behavioral costs, retail "lottery ticket" demand, and institutional constraints. A structural change in these forces would represent genuine mechanism risk.

In practice, which would require some combination of:

- **Behavioral shifts**
Professional allocators become less benchmark- and quarter-focused and can tolerate multi-year tracking error; retail investors stop chasing speculative, positively skewed securities like junior miners, distressed services, and leveraged REITs; loss-chasing and doubling-down behavior fades.
- **Institutional changes**
Long-only mandates gain broad flexibility to use leverage and shorts; pensions and other large pools of capital can freely run structural hedges; compensation and governance frameworks rely much less on benchmark-relative performance.
- **Business model reforms**
Fragile sectors permanently improve: junior miners adopt disciplined capital allocation and higher exploration success rates; oilfield services gain durable pricing power; mREITs operate at materially lower leverage.

These are theoretically possible but collectively unlikely. They would require coordinated shifts in investor psychology, regulation, compensation, and industry structure. So far, the trend has been in the opposite direction: more frequent performance measurement, tighter career risk, and persistent demand for speculative exposures.

8.4 Expected Behavior vs Actual Failure

It is important to distinguish between outcomes that are part of the design and those that would indicate genuine failure. Expected behavior includes: underperformance or negative performance during strong bull markets, flat performance in quiet periods, and limited benefits in small (5–10%) pullbacks. These are the ongoing "costs" of holding structural crisis protection.

Actual failure would look different: a true economic crisis with widening credit spreads and flight-to-quality flows in which SCI does not generate positive returns; a sustained breakdown in the link between fragile business models and crisis losses; or clear evidence that the behavioral, retail, and institutional mechanisms described above have structurally reversed. Allocators should expect and tolerate the former while monitoring for the latter.

9. Conclusion

This paper has argued that crisis protection need not come through actively managed strategies or regime-dependent diversifiers. SCI demonstrates an alternative: exploiting

persistent structural relationships between durable assets and fragile business models that widen predictably during economic stress.

The evidence presented—spanning crisis episodes, robustness testing, and portfolio integration—supports the core thesis: the strategy's edge is **structural rather than statistical**. It is embedded in business model architecture and institutional constraints rather than transient market inefficiencies.

The 2022 inflation shock, where both stocks and bonds declined simultaneously yet SCI generated positive returns (+24.2%), provides particularly strong validation that the mechanism is regime independent.

For allocators capable of maintaining discipline across full market cycles—viewing the strategy as permanent portfolio infrastructure rather than tactical allocation—SCI offers a rare combination: meaningful crisis protection derived from structural market mechanics rather than expensive derivative overlays or regime-dependent diversifiers.

The greatest challenge is not whether the mechanism will work but whether investors will hold it. The risk is that investors allocate at exactly the wrong times—after crises when forward returns are weakest—and abandon the strategy during bull markets before the next crisis occurs.

The strategy exists in a market niche that remains structurally under-allocated: the gap between benchmark-constrained institutional investors (who cannot short or use leverage) and behavior-driven retail investors (who chase lottery tickets).

Those willing to occupy this uncomfortable middle ground—enduring the behavioral cost of lagging rallies in exchange for compounding through crises—can access protection that market structure ensures will remain available.

10. Legal Disclaimers

10.1 Performance Presentation and Limitations

All index values and returns in this paper are hypothetical and presented solely for research and illustration. They do not reflect actual trading of capital. Past simulated performance is not indicative of future results; live results will differ, potentially materially.

Hypothetical results have inherent limitations. They do not reflect actual trading, may be prepared with the benefit of hindsight, and do not capture the impact of liquidity, short availability, margin calls, execution constraints, or behavioral decision-making under stress. No representation is being made that any investor will or is likely to achieve profits or losses similar to those shown.

All figures are shown gross of any practical implementation costs. The simulations exclude management and performance fees, transaction costs (bid–ask spreads, market impact, commissions), financing costs for leverage and short positions, taxes, and other operating expenses. Any investable product referencing this index would experience lower returns than the index itself, and the impact could be meaningful over time. For an estimation of implementation costs and capacity limits, see Appendix E.

The index assumes perfect liquidity and execution at closing prices with no slippage. Real–world implementation would face liquidity constraints, execution delays, and market impact that are not captured here. For GDXJ, launched in 2009, earlier history is proxied using GDX. The 2007–2025 period reflects the collective period of existence for the constituent ETFs; no pre–2007 data is available.

Because implementation choices (vehicle structure, financing terms, borrow costs, fee schedule, tax regime) vary substantially by investor, this paper does not attempt to estimate net–of–cost performance. Those considerations are best evaluated in the context of a specific implementation.

10.2 Index Status

SCI is a rule–based benchmark index, not an investable product, investment fund, or model portfolio. The index cannot be purchased or traded directly. Daily calculation is performed independently by IndexOne, a third–party index calculation agent, based on a codified ruleset that is publicly available. SCI has been calculated and published daily since November 2025 ([link](#)).

10.3 Data Sources and Verification

All performance metrics, correlation statistics, and index–level analytics presented in this paper are derived from the IndexOne historical data series, which represents the retroactive application of the codified ruleset. The index calculation uses total return series for the underlying ETFs, incorporating all dividends and corporate actions.

For granular pair–level analysis and parameter sensitivity testing, historical daily ETF data was sourced from Norgate Data. Norgate series are adjusted for dividends, splits, and corporate actions to ensure accurate total return representation when testing alternative weighting schemas.

BTOP50 monthly data was sourced from BarclayHedge.³ The BTOP50 Index seeks to replicate the overall composition of the managed futures industry.

10.4 General Disclosures

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³ BarclayHedge, *Barclay BTOP50 Index (BTOP50)* — monthly index returns downloaded from the BarclayHedge indices portal, accessed December 15, 2025 ([link](#)).

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Appendix A: Credit Pair Deep Dive

Executive Summary

The Credit Pair is a long–short trade designed to generate positive returns during periods of market stress. The strategy pairs a long position in intermediate Treasury bonds (IEI at 80% allocation) against a short position in mortgage REITs (REM at 40% allocation), creating structural convexity that profits when credit spreads widen and flight–to–quality flows occur. Over the 18–year backtest period (2007–2025), the strategy has exhibited exceptional crisis alpha, generating substantial positive returns during every major equity market drawdown exceeding 15%. The strategy exhibits a –64% correlation to SPY and captures –56% of SPY downside in down SPY months (converting equity losses into strategy gains), making it a powerful portfolio diversifier and tail hedge.

Metric	Value ⁴
Backtest Period	2007–2025 (18-years)
CAGR	0.58%
Annualized Volatility	12.1%
Sharpe Ratio	0.09
Maximum Drawdown	–40%
Correlation to SPY	–0.64
Downside Capture (vs. SPY)	–56%
Upside Capture (vs. SPY)	–28%

Convexity Mechanism

The strategy profits from the divergence between Treasury bonds and mortgage REITs during stress events:

- **Flight to Quality:** During market stress, investors flee risky assets and buy Treasuries. IEI rises as yields fall, contributing positive returns to the long leg.
- **Credit Spread Widening:** Mortgage REITs suffer from widening credit spreads, increased financing costs, and mark–to–market losses on their leveraged portfolios. REM falls, contributing positive returns to the short leg.
- **Leverage Amplification:** Mortgage REITs typically operate with 6–10x leverage, amplifying losses during stress periods and accelerating the divergence from Treasury bonds.

⁴ All performance results included in this paper are hypothetical based on backtested results and do not guarantee future results. See important disclosures in Section 10.

Robustness: The Credit Spread Mechanism

A natural question is whether the strategy performance is specific to REM or reflects a broader, more durable phenomenon. Testing confirms the latter: the same long–short structure generates crisis convexity using alternative short instruments including MORT (VanEck Mortgage REIT Income ETF), HYG (iShares High Yield Corporate Bond ETF), and JNK (SPDR Bloomberg High Yield Bond ETF). The high yield implementations produce lower convexity per dollar of exposure than the mREIT ETFs, but all exhibit the same directional behavior during stress. This validates that the return driver is credit spread behavior generally, not idiosyncratic to any single instrument.

Why Credit Spreads Create Structural Convexity

The strategy exploits a fundamental asymmetry in credit spread behavior:

- **Bounded Compression:** Credit spreads cannot go negative and have a practical floor near historical tights. The upside for the short leg (spreads tightening further) is inherently limited.
- **Unbounded Expansion:** There is no theoretical ceiling on spread widening during stress. When fear dominates, spreads can expand to multiples of their normal levels.
- **Asymmetric Velocity:** Spread compression occurs gradually over years as confidence builds. Spread expansion occurs violently over weeks or months when risk appetite collapses.

This asymmetry means the short credit leg has a positively skewed payoff profile: limited losses during calm periods, outsized gains during stress. The strategy systematically harvests this skew.

Why mREIT ETFs Over High Yield?

REM and MORT are the preferred implementations because mortgage REIT leverage (typically 6–10x) amplifies the credit spread dynamic, generating more convexity per dollar of short exposure. During stress, mREITs face a triple headwind: spread widening on their assets, mark–to–market losses on leveraged portfolios, and potential funding stress as repo markets tighten. This creates a more powerful divergence from Treasuries than unleveraged high yield bonds. The HYG and JNK validation demonstrates that the strategy is not dependent on mREIT–specific factors. The core mechanism—credit spread asymmetry—is a durable feature of fixed income markets that has persisted across decades and market regimes.

Parameter Robustness: Short Weight Sensitivity

A critical question for any systematic strategy is whether results are sensitive to specific parameter choices. To test robustness, we varied the short REM weight from 20% to 60% (versus the baseline 40%) while holding the long IEI position constant at 80%:

The results reveal an important tradeoff inherent in the Credit Pair construction. Lower short weights (20–30%) produce higher CAGR and lower volatility—but provide less crisis protection during severe drawdowns. Higher short weights (50–60%) maximize

crisis convexity but introduce negative CAGR that erodes long-term returns. The baseline 40% short weight represents a balanced choice: it sacrifices some steady-state return to maintain meaningful crisis protection. Critically, the strategy remains profitable across all configurations from 20% to 45%, demonstrating that results are not dependent on precise parameter optimization.

Short Weight	CAGR	Volatility	Max Drawdown
20%	1.60%	6.11%	-25.3%
25%	1.37%	7.28%	-29.0%
30%	1.13%	8.50%	-32.6%
35%	0.87%	9.75%	-36.2%
40% (baseline)	0.58%	11.03%	-40.1%
45%	0.28%	12.33%	-46.0%
50%	-0.04%	13.63%	-51.6%
55%	-0.38%	14.95%	-56.7%
60%	-0.74%	16.27%	-61.4%

Crisis Performance Analysis

The strategy has been tested across five major equity market drawdowns exceeding 15% from local peaks. In each instance, the Credit Pair generated positive returns while SPY experienced significant losses.

The strategy performance during the 2008 Global Financial Crisis (+42.5% vs. SPY -55.2%) and the COVID-19 crash (+47.1% vs. SPY -33.7%) demonstrates the structural nature of the convexity—returns are not dependent on timing or discretionary decisions but emerge automatically from the asset dynamics during stress.

Notably, the 2022 bear market provides critical validation that the Credit Pair is not simply a duration bet. In 2022, rising rates devastated traditional bond portfolios—TLT fell over 30% and the 60/40 portfolio suffered its worst year in decades. Yet the Credit Pair returned +5.6% because the short REM leg generated gains that more than offset losses on IEL. The strategy convexity comes from credit spread dynamics, not interest rate exposure.

Crisis Period	SPY Return	Credit Pair Return
GFC (Oct 2007–Mar 2009)	–55.2%	+42.5%
Q4 2018 Selloff	–19.3%	+5.8%
COVID–19 (Feb–Mar 2020)	–33.7%	+47.1%
2022 Bear Market	–24.5%	+5.6%
2025 Tariff Correction	–18.8%	+8.6%

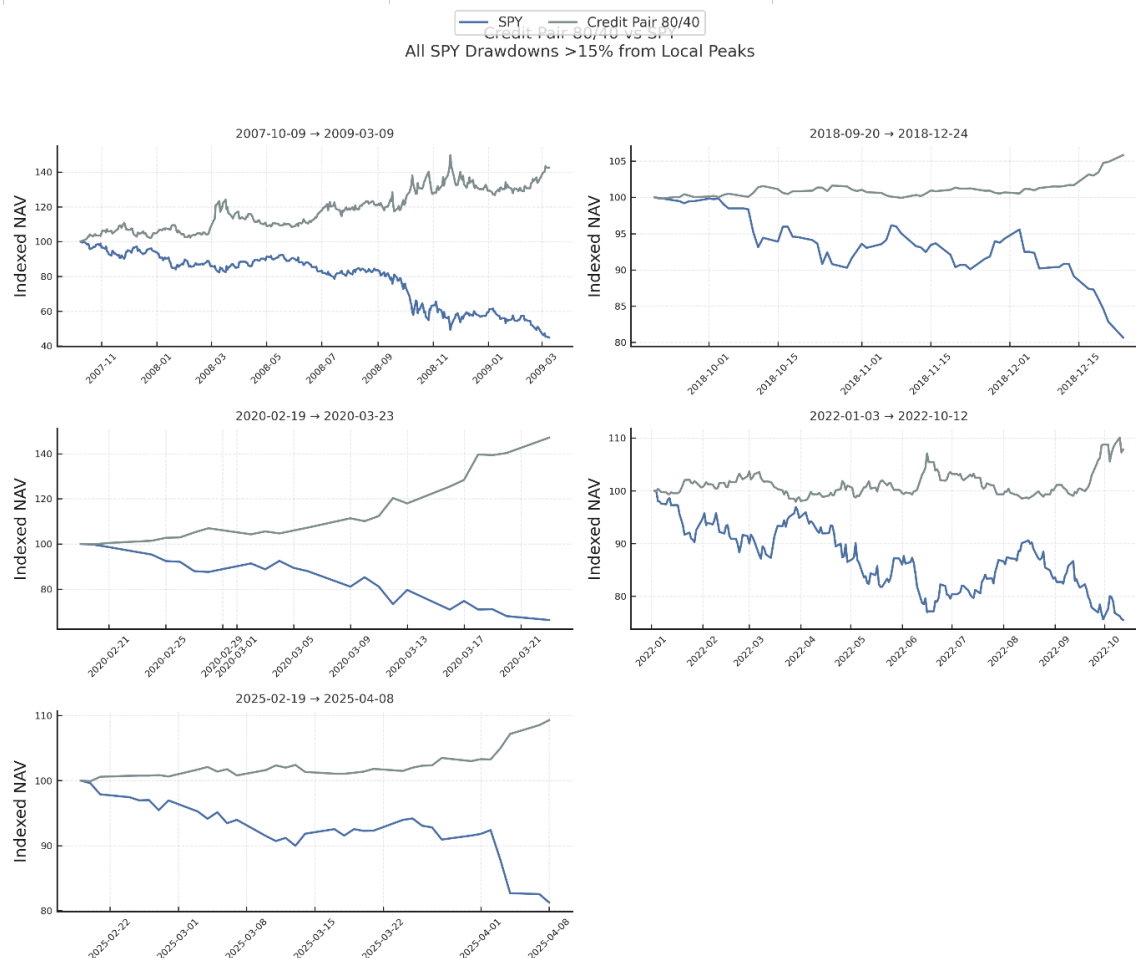


Figure 1: Credit Pair performance during major equity drawdowns shows consistent positive returns when SPY experienced losses greater than 15%.

Interest Rate Sensitivity

A key consideration for any Treasury–based strategy is interest rate sensitivity. Analysis of the Credit Pair behavior during significant TLT (iShares 20+ Year Treasury Bond ETF) moves reveals an asymmetric profile: the strategy exhibits positive average returns during months when TLT rises 4%+ (flight–to–quality environments) and

approximately flat returns during months when TLT falls 4%+ (rising rate environments). This asymmetry occurs because credit stress typically accompanies Treasury rallies, while rising rate environments often reflect healthy economic conditions where the short REM leg partially offsets IEI losses.

TLT Monthly Shock	Avg Credit Pair Return
TLT Up $\geq 4\%$ (rates falling)	+1.77%
TLT Down $\leq -4\%$ (rates rising)	~0%

Capture Ratio Analysis

The strategy value proposition is quantified through its capture ratios relative to SPY:

The –28% upside capture indicates the strategy gives back approximately 0.28% for every 1% SPY gains during bull markets—the cost for crisis protection. The –56% downside capture means the strategy generates approximately 0.56% for every 1% SPY losses during corrections.

Capture Metric	Value
Upside Capture (SPY up months)	–29%
Downside Capture (SPY down months)	–56%
Correlation to SPY	–0.64

Risk Factors

- Extended Bull Markets: During prolonged equity bull markets without corrections, the strategy will experience gradual drawdowns as mortgage REITs outperform Treasuries. The –40% maximum drawdown (2020–2025) reflects this dynamic.

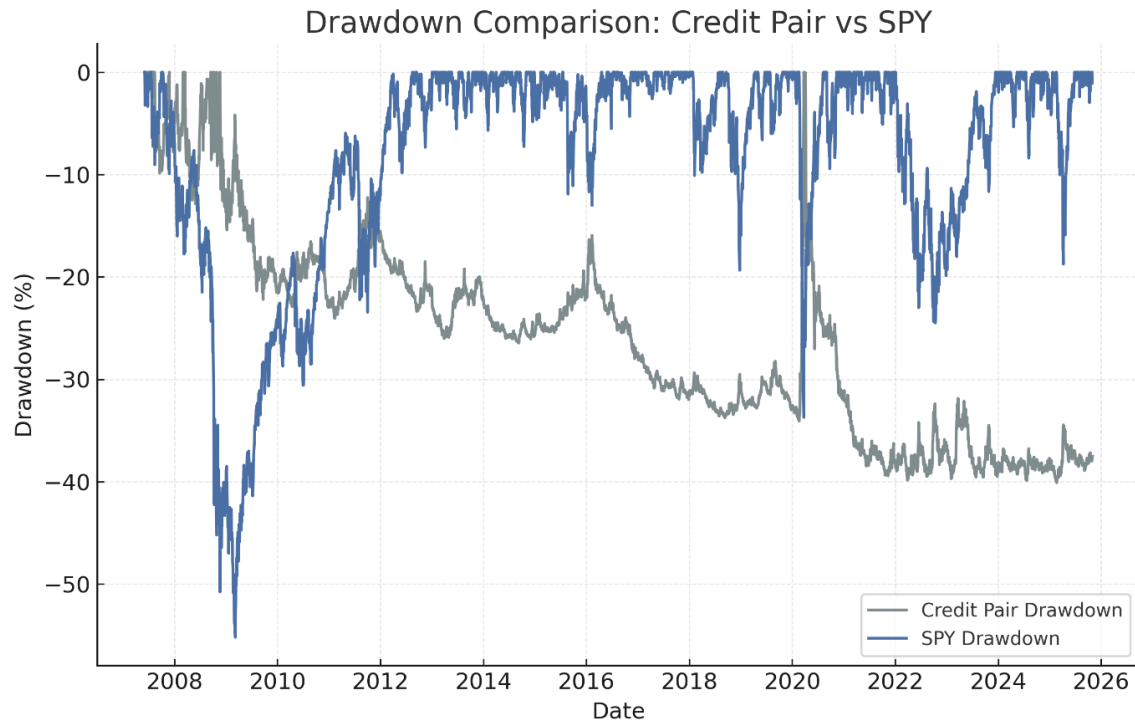


Figure 2: Drawdown comparison reveals Credit Pair's inverse relationship to equities, with its deepest drawdowns occurring during sustained equity bull markets.

- **Stagflation Scenario:** A rising rate environment combined with credit stress could pressure both legs—Treasuries falling while mortgage REITs also decline. Historical analysis suggests this scenario is less damaging than pure duration exposure but remains a consideration.

Appendix B: Gold Pair Deep Dive

Executive Summary

Within SCI, the Gold Pair represents 40% of the net long position. The strategy pairs a long position in physical gold (GLD at 60% allocation) against a short position in junior gold miners (GDXJ at 20% allocation), creating structural convexity that profits from the divergence between physical gold and junior gold mining equities during flight-to-safety events. Over the 18-year backtest period (2007–2025), the strategy achieved 3.60% CAGR with only 6.99% volatility. The strategy exhibits negative correlation to SPY (–0.26) and captures –23% of SPY downside (converting equity losses into strategy gains), making it a powerful crisis hedge that activates precisely when needed most.

Key Performance Metrics

Metric	Value ⁵
Backtest Period	2007–2025 (18-years)
CAGR	3.60%
Annualized Volatility	6.99%
Sharpe Ratio	0.64
Sortino Ratio	1.09
Maximum Drawdown	–17.6%
Correlation to SPY	–0.26
Downside Capture (vs. SPY)	–23%
Upside Capture (vs. SPY)	~0%

Convexity Mechanism

The strategy profits from the structural divergence between physical gold and junior gold miners during market stress events:

- **Flight-to-Quality Dynamics:** During market crises, physical gold receives safe-haven bids while junior miners—as equity instruments with operational leverage—face selling pressure. Gold represents monetary stability; junior miners represent speculative risk. When fear rises, capital rotates from complexity to simplicity, from fragility to durability.

⁵ All performance results included in this paper are hypothetical based on backtested results and do not guarantee future results. See important disclosures in Section 10.

- **Operating Leverage Amplification:** Junior gold miners exhibit 2–3x leveraged responses to gold prices due to their high fixed costs and marginal economics. When gold prices fall, mining margins collapse. But even when gold prices rise during crises, junior miners often fall anyway due to broader equity market contagion and funding market stress.
- **Funding Market Vulnerability:** Junior miners consume capital for years before generating revenue, requiring constant access to equity markets. During crises, funding markets freeze. Unable to raise capital, juniors cut exploration budgets, face dilutive emergency financing, or approach insolvency—destroying equity value regardless of gold price direction.
- **Behavioral Bid Collapse:** Retail investors treat junior miners as lottery tickets, attracted by low share prices and potential for extreme gains. Research shows these investors systematically overpay for lottery-like stocks. During crises, risk appetite vanishes, the promotional machine stops, and junior miners lose their natural bid precisely when they need it most.

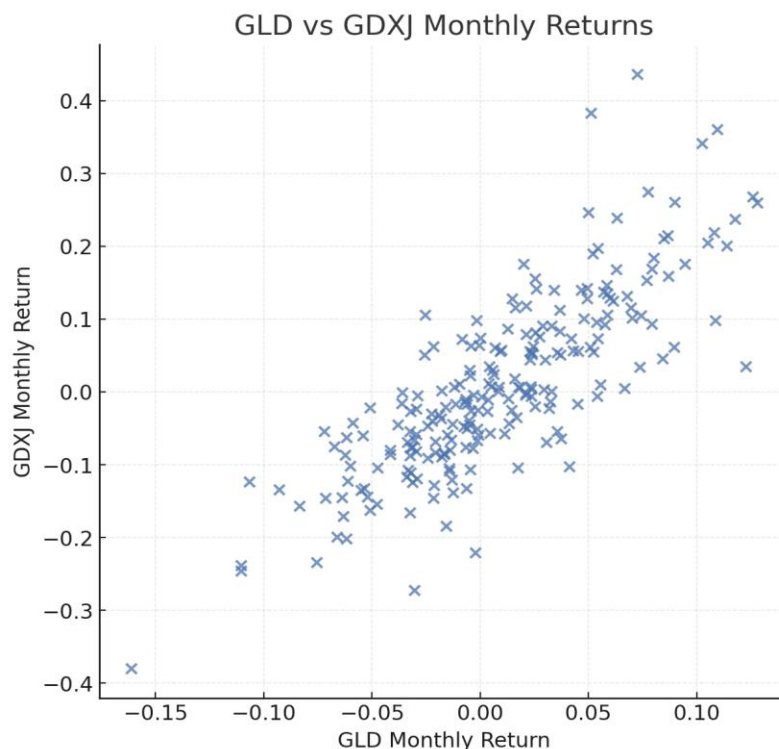


Figure 1: Scatter plot of GLD vs GDXJ monthly returns demonstrates asymmetric beta relationship with miners amplifying losses but not gains

The scatter plot illustrates GDXJ asymmetric beta to GLD. While both assets share directional gold exposure, GDXJ exhibits significantly greater volatility and more extreme moves in both directions. When GLD falls 10%, GDXJ often falls 20–30%. When GLD rises 10%, GDXJ gains vary wildly from 5% to 40%. This asymmetry—

combined with GDXJ structural underperformance over time—is the source of the strategy convexity and positive carry.

Robustness: The Structural Unprofitability of Junior Gold Miners

A natural question is whether the strategy performance is specific to the backtest period or reflects a durable structural phenomenon. Analysis of the junior gold mining sector reveals deep structural reasons why GDXJ systematically underperforms GLD over time—reasons documented extensively in academic research and unlikely to change.

Harsh Exploration Economics

Junior gold mining companies face extraordinary odds. A miniscule number of exploration prospects develop into operational mines. The journey from initial prospect to producing mine typically spans 10–20 years and requires hundreds of millions of dollars in capital.

The Downward Dilution Spiral

Junior miners face a structural challenge distinguishing them from virtually every other business model: they consume capital for years before generating any revenue. Unable to finance operations through cash flow, juniors must repeatedly access capital markets through equity issuances that dilute existing shareholders. In addition, a significant portion of capital raised by junior explorers is spent on corporate overhead rather than exploration—the ostensible purpose of these companies.

Empirical Evidence

Recent academic research provides additional validation of this structural underperformance. Baur, Trench, and Tay (2025) document that the VanEck Gold Miners ETF (GDX) underperformed gold bullion (GLD) by approximately 350% over the 2006–2025 period—equivalent to –6.5% annual underperformance. Their analysis of 62 individual gold mining companies found that 38 underperformed gold while only 24 outperformed, with median returns trailing gold even after including dividends.

The authors attribute this persistent underperformance to the finite life of mines: Even if mining companies were not exposed to risks associated with mining such as production imposts due to operational challenges, weather, environmental regulations, country risks and so forth, gold mining companies would still underperform gold bullion on average in the long run.

Crucially, their regression analysis reveals that during equity market crises, gold miners decouple from gold and recouple with the equity market—precisely the dynamic that generates crisis alpha in the Gold Pair.

The same study quantifies this asymmetry: gold miners exhibit a gold beta of approximately 1.7–1.8 combined with a market beta of 0.4–0.5, confirming miners amplify both gold and equity market movements. Their GARCH analysis further reveals that gold exhibits an inverse leverage effect (volatility rises on positive shocks, consistent with safe-haven behavior), while gold miners exhibit a positive leverage effect typical of levered equities. This asymmetric volatility response is precisely why the Gold Pair

generates crisis alpha: when markets stress, gold volatility rises with prices while miner volatility rises with losses.

Behavioral Persistence

The central puzzle is not that these investments destroy value, but that retail investors continue to fund the sector despite overwhelming evidence of failure. Academic research documents that retail investors exhibit strong preferences for lottery-like stocks: low prices, high volatility, positive skewness, and potential for extreme positive returns. Junior miners possess precisely these characteristics. Loss-chasing behavior compounds the problem: investors experiencing large unrealized losses become significantly more likely to invest in lottery stocks, perpetuating the cycle.



Figure 2: Cumulative NAV comparison shows GLD appreciating 6-fold while GDXJ remains underwater after 18-years

GLD maximum drawdown was -45% while GDXJ suffered an -85% peak-to-trough decline, asymmetry that forms the foundation of the Gold Pair convexity.

Parameter Robustness: Short Weight Sensitivity

To test robustness, we varied the short GDXJ weight from 16% to 24% (versus the baseline 20%) while holding the long GLD position constant at 60%:

The results demonstrate stability across the parameter range. All configurations produce positive Sharpe ratios between 0.52 and 0.71, with maximum drawdowns contained between -17.2% and -20.6% . The relationship is intuitive: lower short weights (16–18%) produce higher returns with slightly less crisis protection, while higher short weights (22–24%) sacrifice some return for additional hedging. The baseline 20% short weight represents a balanced choice, but the strategy remains attractive across the entire range tested; evidence that results are not dependent on precise parameter optimization.

Short Weight	Net Exposure	CAGR	Sharpe	Max DD
16%	44%	4.20%	0.71	-17.4%
17%	43%	4.05%	0.70	-17.3%
18%	42%	3.90%	0.68	-17.2%
19%	41%	3.75%	0.66	-17.3%
20% (baseline)	40%	3.60%	0.64	-17.6%
21%	39%	3.44%	0.61	-18.0%
22%	38%	3.28%	0.58	-18.8%
23%	37%	3.12%	0.55	-19.7%
24%	36%	2.96%	0.52	-20.6%

Crisis Performance Analysis

The strategy has been tested across five major equity market drawdowns exceeding 15% from local peaks. The Gold Pair was positive in each of the five drawdowns examined in this paper, consistent with the flight-to-quality mechanism in this sample.

The Global Financial Crisis performance is particularly noteworthy: while SPY declined 55% and even passive gold exposure (40% GLD) gained only 11%, the Gold Pair delivered +17.7%—capturing the flight-to-quality premium as capital rotated from mining equities to physical gold. During COVID-19, when equity markets collapsed 34% in just five weeks, the Gold Pair gained 4.4% while passive gold exposure posted a small loss.

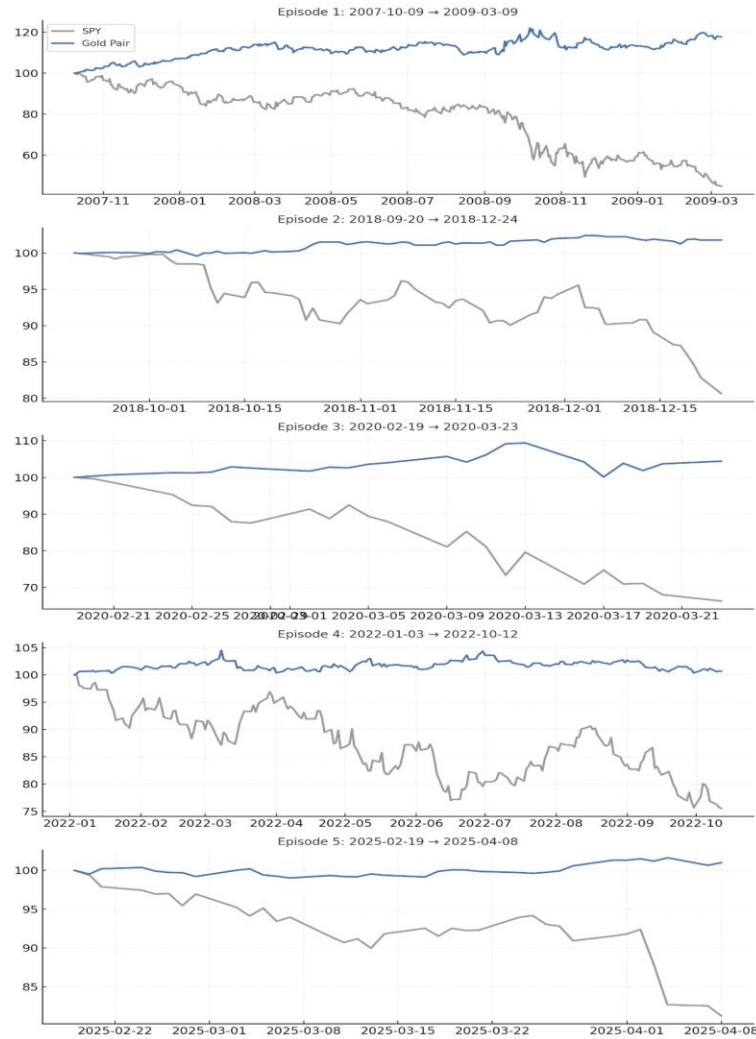


Figure 3: Gold Pair performance during major equity drawdowns demonstrates negative correlation to SPY with positive returns in most crisis periods.

Crisis Period	SPY Return	GLD Return	Gold Pair Return
GFC (Oct 2007–Mar 2009)	–55.2%	+10.7%	+17.7%
COVID–19 (Feb–Mar 2020)	–33.7%	–1.3%	+4.4%
2022 Bear Market	–24.5%	–2.8%	+0.6%
Q4 2018 Selloff	–19.3%	+2.0%	+1.8%
2025 Correction	–18.8%	+0.7%	+1.3%

Drawdown Displacement: Shifting Pain to Where It Is Tolerable

Beyond absolute drawdown reduction (−17.6% vs. −18.2% for 40% GLD), the Gold Pair exhibits a critical behavioral advantage: its worst drawdowns occur during different market regimes than the benchmark's worst drawdowns.

The 40% GLD allocation maximum drawdown (−18.2%) occurred during the 2011–2015 gold bear market—a period of falling gold prices that coincided with equity market weakness (SPY bottomed in January 2016) which would have compounded portfolio level pain. The Gold Pair maximum drawdown (−17.6%) occurred when SPY was rallying, which would have been much easier to tolerate at the portfolio level. The short GDXJ leg provides the most value precisely during equity market stress when investors need protection most and then recharges during equity bull markets.



Figure 4: Drawdown comparison shows Gold Pair experiences drawdowns at different times than gold

Equity Regime Analysis

Analysis of the Gold Pair returns across different SPY monthly regimes reveals its true crisis alpha characteristic: the strategy generates its strongest returns during equity selloffs.

The strategy delivers +0.27% during normal markets (SPY between −5% and +5%), +1.47% during equity selloffs (SPY down more than 5%), and −0.33% during equity rallies (SPY up more than 5%). This crisis-activated return profile is precisely what a tail hedge should deliver: modest positive drift during normal markets, significant positive returns during stress, and acceptable underperformance during euphoric rallies when protection is not needed.

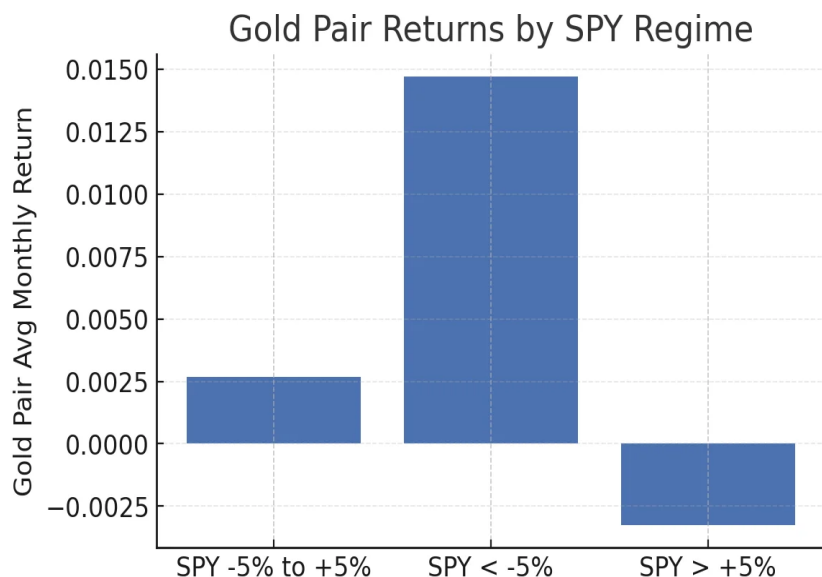


Figure 5: Gold Pair monthly returns by SPY regime show strongest performance during severe equity selloffs while remaining flat to slightly negative in bull markets.

Gold Sector Sensitivity

A key consideration is how the Gold Pair behaves during large gold price moves:

The strategy exhibits dampened sensitivity to gold price moves. During months when GLD rises 4%+, the Gold Pair captures approximately 20% of the move (+1.42% vs. +7.23%). During months when GLD falls 4%+, the strategy captures only 21% of the decline (-1.43% vs. -6.92%). This symmetric dampening reflects the strategy crisis-focused design: it sacrifices gold bull market upside to fund crisis protection.

GLD Monthly Move	Avg GLD Return	Avg Gold Pair Return
GLD Up $\geq 4\%$	+7.23%	+1.42%
GLD Down $\leq -4\%$	-6.92%	-1.43%

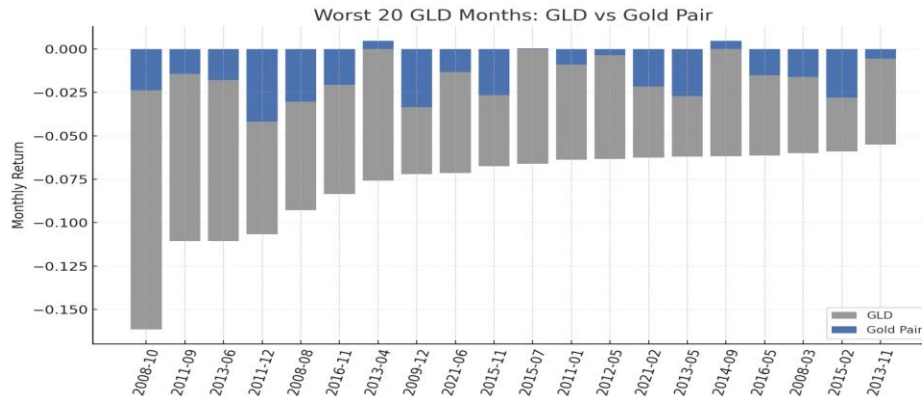


Figure 6: Gold Pair performance during GLD's worst months demonstrates significant outperformance through the short GDXJ position, which harvests losses in junior miners.

The chart shows the Gold Pair systematic cushioning of gold worst months. In nearly every instance of severe GLD decline, the Gold Pair remains near zero while GLD plunges. During the worst month (October 2008 at -16%), the Gold Pair loss was contained to just -2% . This consistent downside protection is the practical manifestation of the strategy's structural convexity.

Capture Ratio Analysis

The strategy value proposition is quantified through its capture ratios relative to SPY. The negative correlation and beta indicate the strategy provides genuine diversification benefit. The -23% downside capture means the strategy generates positive returns when equities fall—a rare characteristic that provides crisis protection without the negative carry of traditional tail hedges. The near-zero upside capture indicates the strategy operates independently during equity rallies.

Metric	Value
Upside Capture (SPY up months)	$\sim 0\%$
Downside Capture (SPY down months)	-23%
Correlation to SPY	-0.26
Beta to SPY	-0.10

Comparison to Passive Gold Allocation

The appropriate benchmark for the Gold Pair is a passive 40% GLD allocation—matching the strategy for net gold exposure:

The Gold Pair delivers comparable returns (3.60% vs. 4.00% CAGR) with meaningfully better risk-adjusted performance: 10% higher Sharpe ratio (0.64 vs. 0.58), 12% higher Sortino ratio (1.09 vs. 0.97), and slightly lower maximum drawdown (-17.6% vs. -18.2%). The Ulcer Index, which measures both depth and duration of drawdowns—shows the Gold Pair at 7.0 vs. 8.9 for passive gold, indicating a 21% reduction in

drawdown pain. Perhaps most importantly, the correlation to SPY shifts from neutral (0.02) to meaningfully negative (-0.26), transforming gold from a non-correlated asset into an active crisis hedge.

Metric	Gold Pair	40% GLD
CAGR	3.60%	4.00%
Volatility	6.99%	6.88%
Sharpe Ratio	0.64	0.58
Sortino Ratio	1.09	0.97
Maximum Drawdown	-17.6%	-18.2%
Ulcer Index	7.0	8.9
% Positive Months	58.2%	53.2%
Correlation vs SPY	-0.26	0.02

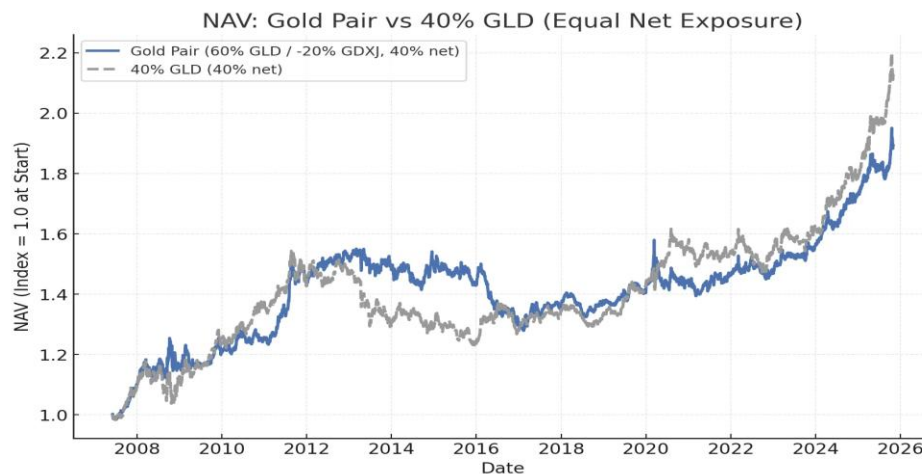


Figure 7: Cumulative NAV growth comparison shows Gold Pair delivering similar absolute returns to 40% GLD with reduced volatility.

The NAV comparison shows both strategies reaching similar endpoints ($\sim 2.0x$), but with different paths. The Gold Pair (blue) exhibits a steadier trajectory with shallower drawdowns, while 40% GLD (gray) shows more volatility. The strategies diverge during crisis periods, where the Gold Pair pulls ahead, then reconverge during gold bull markets when the passive allocation catches up.



Figure 8: Rolling 12-month returns demonstrate Gold Pair's consistent volatility reduction relative to GLD while maintaining positive long-term performance.

The rolling 12-month return chart illustrates the volatility compression over investable horizons. GLD (light blue) swings widely from +60% to -30% on a trailing year basis. The Gold Pair (dark blue) exhibits a dramatically narrower band, rarely exceeding $\pm 20\%$. For allocators concerned about client experience and behavioral risk, this smoother return profile reduces the likelihood of panic selling during gold bear markets.

Risk Factors

- **Gold Bull Markets:** During sustained gold rallies, the strategy will significantly underperform passive GLD exposure as the short GDXJ leg generates losses. The strategy captures only ~20% of upside moves in gold.
- **Junior Miner Outperformance Regime:** If junior miners enter a sustained period of outperformance (major discovery cycle, extended gold bull market with risk-on sentiment, or consolidation wave), the short GDXJ leg will generate losses that may not be offset by GLD gains.

Appendix C: Energy Pair Deep Dive

Executive Summary

Within SCI, the Energy Pair represents 20% of the net long position. The strategy pairs a long position in integrated energy majors (XLE at 60% allocation) against a short position in oilfield services (IEZ at 40% allocation), creating structural convexity that profits from the divergence between producers and service providers during sector dislocations. Over the 18-year backtest period (2007–2025), the strategy achieved 3.47% CAGR with only 7.09% volatility—more than doubling the returns of a passive 20% XLE allocation while cutting maximum drawdown by nearly 25%. The strategy exhibits essentially zero correlation to SPY (0.01) and captures –5% of SPY downside (converting equity losses into strategy gains), making it a powerful portfolio diversifier that operates independently of market direction.

Key Performance Metrics

Metric	Value ⁶
Backtest Period	2007–2025 (18-years)
CAGR	3.47%
Annualized Volatility	7.09%
Sharpe Ratio	0.63
Sortino Ratio	1.07
Maximum Drawdown	–15.0%
Correlation to SPY	0.01
Downside Capture (vs. SPY)	–5%
Upside Capture (vs. SPY)	11%

Convexity Mechanism

The strategy profits from the structural divergence between integrated energy majors and oilfield services companies during sector stress events:

- **Capital Structure Divergence:** Integrated majors maintain fortress balance sheets, diversified revenue streams, and significant dividend support. Oilfield services companies operate with higher leverage, concentrated revenue exposure to

⁶ All performance results included in this paper are hypothetical based on backtested results and do not guarantee future results. See important disclosures in Section 10.

E&P capital expenditure budgets, and minimal dividend protection. During commodity downturns, this capital structure divergence amplifies.

- **Cyclical Amplification:** When oil prices decline, E&P companies immediately cut capital expenditure budgets—the sole revenue source for oilfield services companies. Services companies face double leverage: commodity price exposure multiplied by activity level cuts. A 30% oil price decline can translate into 50%+ revenue declines for services companies as operators slash drilling programs.
- **Operating Margin Collapse:** Oilfield services companies face extreme margin compression during downturns. Industry analysis documents operating margins collapsing from 15% to under 5% during the 2014–2019 period, with cumulative losses exceeding \$96 billion. The high fixed-cost nature of services businesses (equipment, skilled labor, technology development) creates operating leverage that devastates returns during volume declines.
- **Fragmentation Penalty:** The oilfield services sector remains persistently fragmented despite clear industrial logic for consolidation. This fragmentation prevents the sector from capturing value even during recovery periods, as excess capacity and new entrants perpetually compress margins.

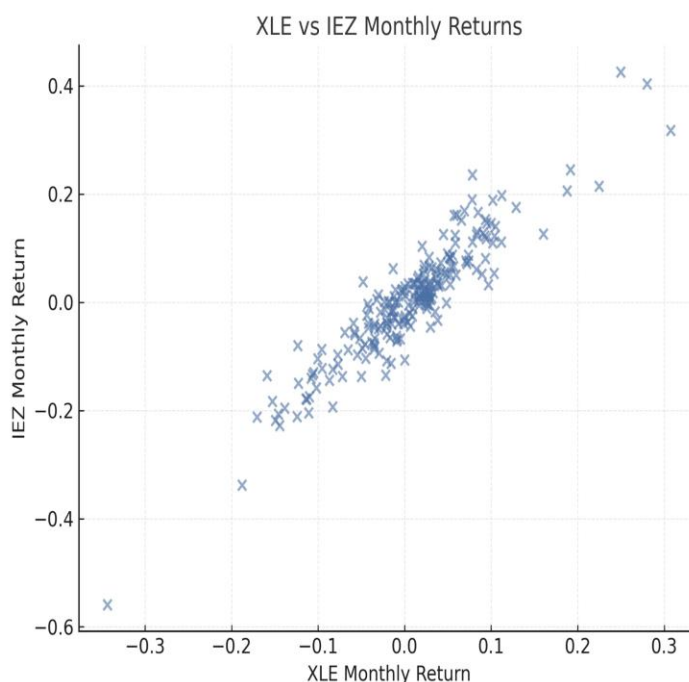


Figure 1: Scatter plot of XLE vs IEZ monthly returns reveals asymmetric beta with services companies amplifying downside losses during energy selloffs.

The scatter plot illustrates IEZ asymmetric beta to XLE. While both assets are highly correlated, IEZ exhibits significantly greater volatility and more extreme downside moves. When XLE falls 15–20%, IEZ often falls 30–40%. This asymmetry—combined with IEZ structural underperformance over time—is the source of the strategy convexity.

Robustness: The Structural Underperformance of Oilfield Services

Oilfield services companies like Schlumberger, Halliburton, and the constituents of IEZ—provide drilling, equipment, and support services to oil and gas producers. Their structural disadvantage stems from the Bullwhip Effect in the energy supply chain: OFS companies act as downstream vendors to E&P capital allocators, bearing the brunt of volatility amplification without the balance sheet protection or asset–ownership moats enjoyed by the integrated majors.

Second–Derivative Exposure

The primary driver of OFS volatility is not the oil price itself, but second–derivative exposure to producer capital expenditure (capex). Integrated majors continually adjust capex to protect their dividends and free cash flow. A 10% drop in crude prices often triggers a disproportionate 30–40% cut in discretionary drilling budgets. OFS revenues are derived entirely from this discretionary capex—when majors cut spending, the bullwhip snaps the OFS sector hard. Result: OFS earnings volatility is typically $2\times$ – $3\times$ that of the underlying commodity. Industry analysis confirms this structural vulnerability. McKinsey (2020) documented that the oil field service and equipment sector delivered –18.7% annualized total shareholder returns from 2014–2019, compared to –2.5% for integrated oil and gas operators—a 16 percentage point annual performance gap.

Renters vs Owners

Feature	Integrated Majors (Owners)	Oil Field Services (Renters)
Asset Base	Own the Resource (reserves appreciate with inflation)	Own the Iron (rigs/equipment depreciate constantly)
Pricing Power	Price Makers (OPEC+ influence, global benchmarks)	Price Takers (competitive bidding, utilization–driven)
Natural Hedges	Downstream profits when crude falls	None—pure upstream exposure
Through–Cycle ROE	10–15%	5–8% (often negative in downturns)
Beta to Market	0.7–1.0 (stable)	1.8–2.5 (amplified)
Capital Cycle Sensitivity	Counter–cyclical (can invest when oil prices are low)	Pro–cyclical (forced to invest when prices are high/equipment expensive)

The Hamster Wheel of Capex

OFS companies suffer from bad capital intensity. They must constantly reinvest heavily in new equipment just to maintain revenue flat. If they stop spending, their fleet becomes obsolete. Majors, conversely, can sit on long-life reserves with minimal spend during downturns, preserving ROE. Every 5–7 years, the OFS sector undergoes massive restructuring and bankruptcy waves, permanently impairing equity capital.

Unlike technology companies that can build durable competitive advantages through R&D, OFS innovation is rapidly commoditized. Companies must continuously reinvest just to maintain competitive parity, with efficiency gains captured by customers through competitive bidding rather than accruing to shareholders. This creates a dynamic whereby OFS companies run faster just to stay in place, while integrated majors harvest the productivity improvements as lower service costs.

Empirical Evidence: The XLE/IEZ Divergence



Figure 2: Cumulative NAV comparison demonstrates XLE appreciation while IEZ experiences persistent structural underperformance

The NAV comparison starkly illustrates the structural underperformance. Over the 18-year period, XLE has delivered approximately 130% cumulative returns while IEZ has lost over 50% of its value. The divergence accelerated dramatically after 2014, as the structural factors outlined above compounded during the commodity downturn. This is not a temporary phenomenon—it reflects the fundamental economics of the oilfield services business model.

Parameter Robustness: Short Weight Sensitivity

To test robustness, we varied the short IEZ weight from 20% to 60% (versus the baseline 40%) while holding the long XLE position constant at 60%:

The results demonstrate stability across the parameter range. All configurations produce positive returns between 2.41% and 3.89% CAGR, with the sweet spot appearing between 30% and 45% short weight where volatility is minimized (7.10–8.49%) and drawdowns are contained (–13.7% to –18.0%). The baseline 40% short weight sits at the inflection point: lower short weights increase volatility due to insufficient hedging, while higher

short weights increase volatility due to excessive short exposure. This natural optimum suggests the strategy construction is well-calibrated rather than over-fitted to historical data.

Short Weight	CAGR	Volatility	Max Drawdown
20%	3.89%	11.19%	-25.4%
25%	3.85%	9.73%	-21.3%
30%	3.76%	8.49%	-17.1%
35%	3.64%	7.57%	-13.7%
40% (baseline)	3.47%	7.10%	-15.0%
45%	3.27%	7.16%	-18.0%
50%	3.02%	7.76%	-21.0%
55%	2.74%	8.77%	-24.3%
60%	2.41%	10.07%	-29.1%

Crisis Performance Analysis

The strategy has been tested across five major equity market drawdowns exceeding 15% from local peaks. The Energy Pair generated positive returns during four of five episodes—and critically, even during the COVID crisis when energy sector experienced its worst month in history, the strategy loss was contained to just 5.3%.

The 2022 bear market is particularly noteworthy: while SPY declined 24%, the energy sector rallied 49% due to the Russia–Ukraine conflict and energy supply disruptions. The Energy Pair captured this upside (+13.9%) while other defensive strategies struggled. This demonstrates the strategy unique characteristic: it provides crisis protection during equity drawdowns while also participating in energy-specific rallies that often occur during inflationary or geopolitical stress.

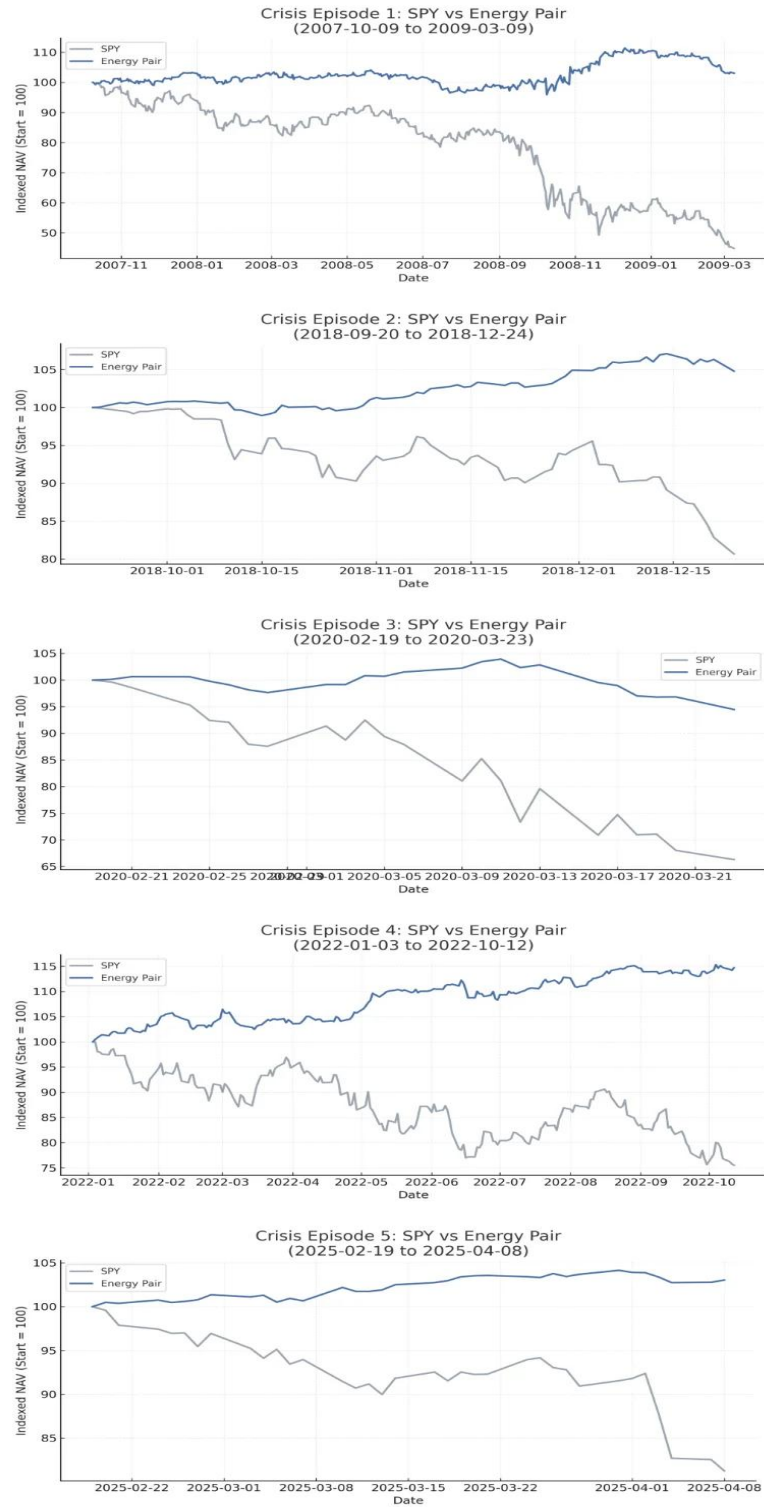


Figure 3: Energy Pair performance during major equity drawdowns shows near-zero correlation to SPY with meaningful positive contribution during 2022 crisis.

Crisis Period	SPY Return	XLE Return	Energy Pair Return
GFC (Oct 2007–Mar 2009)	–54.8%	–46.9%	+3.7%
Q4 2018 Selloff	–18.7%	–27.1%	+4.7%
COVID–19 (Feb–Mar 2020)	–33.4%	–55.5%	–5.3%
2022 Bear Market	–24.1%	+49.0%	+18%
2025 Tariff Correction	–18.6%	–15.6%	+2.7%

Drawdown Displacement: Shifting Pain to Where It Is Tolerable

Beyond absolute drawdown reduction (–15.0% vs. –19.6% for 20% XLE), the Energy Pair exhibits a critical behavioral advantage: its worst drawdowns occur during different market regimes than the benchmark worst drawdowns.

The 20% XLE allocation maximum drawdown (–19.6%) occurred during COVID–19—simultaneously with the equity market worst drawdown, compounding investor pain across the portfolio. In contrast, the Energy Pair maximum drawdown (–15.0%) occurred during a period when SPY was rallying strongly immediately *after* the COVID–19 low, which would have made it more tolerable had it been part of a balanced portfolio. This displacement is not coincidental—the short IEZ leg cushions energy sector drawdowns precisely when they coincide with broader equity stress and then recharges during broader equity market strength.

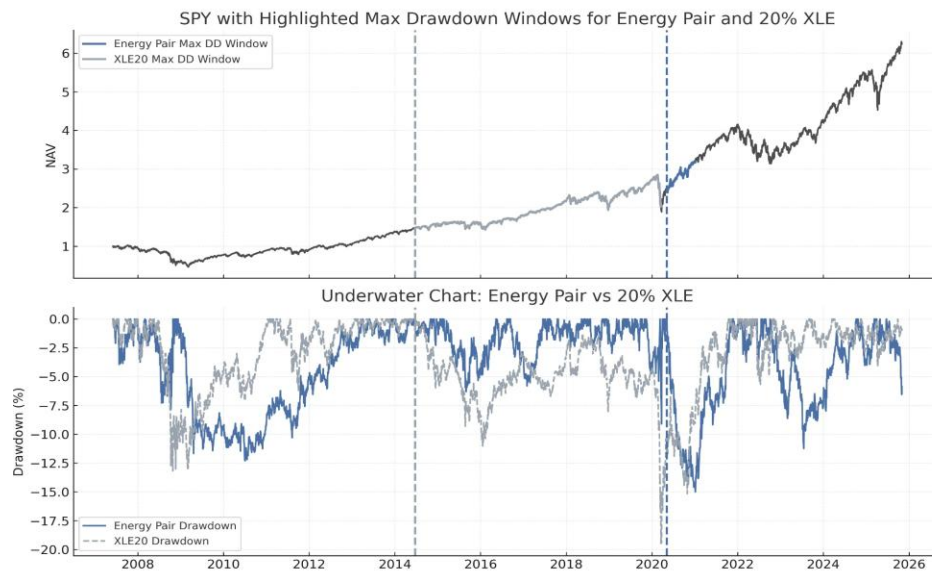


Figure 4: Drawdown comparison reveals different timing for Energy Pair and XLE drawdowns

Equity Regime Analysis

Analysis of the Energy Pair returns across different SPY monthly regimes reveals a striking characteristic: the strategy generates positive returns across all equity market regimes.

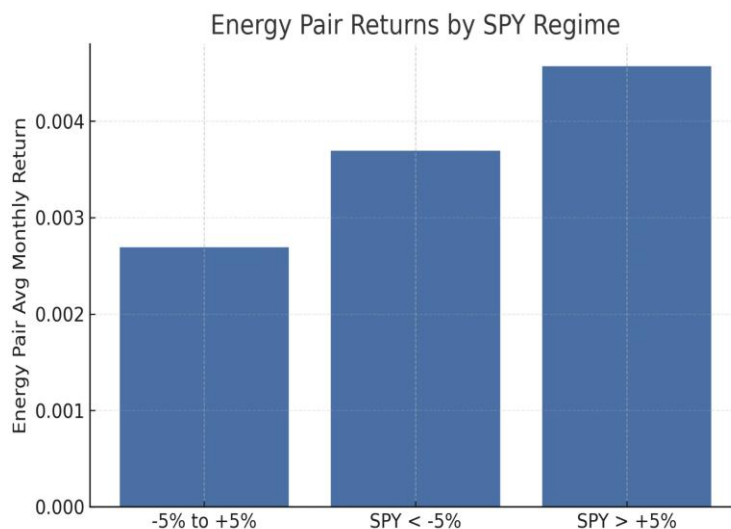


Figure 5: Energy Pair monthly returns by SPY regime demonstrate genuine diversification with near-zero correlation across all equity market conditions.

The strategy delivers positive average returns in all three regimes: +0.27% during normal markets (SPY between -5% and +5%), +0.37% during equity selloffs (SPY down

more than 5%), and +0.46% during equity rallies (SPY up more than 5%). This all-weather characteristic distinguishes the Energy Pair from traditional tail hedges that bleed during normal markets. The near-zero correlation to SPY (0.01) means the strategy operates largely independently of equity market direction.

Energy Sector Sensitivity

A key consideration is how the Energy Pair behaves during large energy sector moves: the strategy exhibits dramatically dampened sensitivity to energy sector moves. During months when XLE rises 4%+, the Energy Pair captures approximately 6% of the move (+0.56% vs. +9.29%). During months when XLE falls 4%+, the strategy captures only 3% of the decline (−0.32% vs. −9.72%). This asymmetry favors the strategy: losses during energy selloffs are minimal, while the structural alpha from the long/short construction generates positive drift over time.

XLE Monthly Move	Avg XLE Return	Avg Energy Pair Return
XLE Up $\geq 4\%$	+9.29%	+0.56%
XLE Down $\leq -4\%$	−9.72%	−0.32%

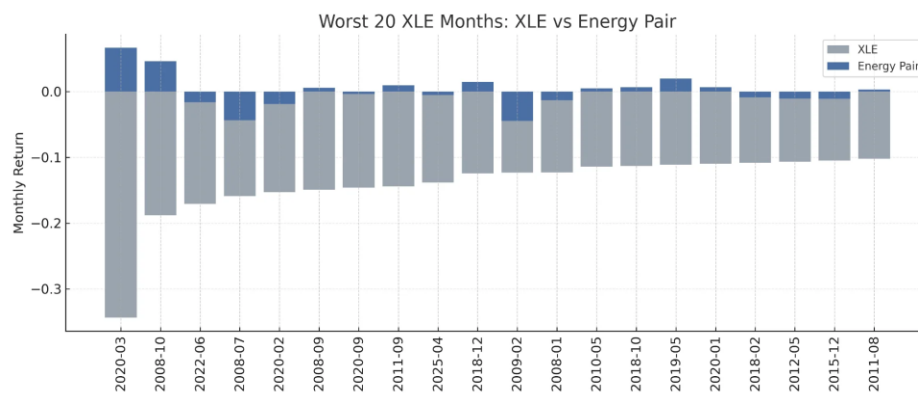


Figure 6: Energy Pair performance during XLE's worst months shows substantial outperformance as the short IEZ position harvests the fragility of services companies.

The chart shows the Energy Pair systematic cushioning of energy worst months. In nearly every instance of severe XLE decline, the Energy Pair remains near zero while XLE plunges. Remarkably, during the two worst months (March 2020 at −34% and October 2008 at −18%), the Energy Pair posted positive returns. This consistent downside protection is the practical manifestation of the strategy structural convexity.

Capture Ratio Analysis

The strategy value proposition is quantified through its capture ratios relative to SPY. The near-zero correlation and beta indicate the strategy operates independently of equity market direction. The −5% downside capture means the strategy generates

modest positive returns during equity selloffs—a rare characteristic that provides genuine portfolio diversification. The 11% upside capture indicates the strategy participates slightly in equity rallies, likely through the correlation between energy stocks and broader markets during risk-on environments.

Metric	Value
Upside Capture (SPY up months)	11%
Downside Capture (SPY down months)	–5%
Correlation to SPY	0.01
Beta to SPY	0.01

Comparison to Passive Energy Allocation

The appropriate benchmark for the Energy Pair is a passive 20% XLE allocation—matching the strategy for net energy exposure:

The Energy Pair delivers dramatically superior returns (3.47% vs. 1.64% CAGR) with better risk-adjusted performance across all metrics: nearly double the Sharpe ratio (0.52 vs. 0.29), more than double the Sortino ratio (1.05 vs. 0.46), and 24% lower maximum drawdown (–15.0% vs. –19.6%). Perhaps most importantly, the correlation to SPY drops from 0.65 for passive energy to essentially zero (0.01) for the Energy Pair—transforming a correlated equity exposure into a true diversifier.

Metric	Energy Pair	20% XLE
CAGR	3.47%	1.64%
Volatility	7.09%	6.02%
Sharpe Ratio	0.52	0.29
Sortino Ratio	1.05	0.46
Maximum Drawdown	–15.0%	–19.6%
Calmar Ratio	0.23	0.08
% Positive Months	57.7%	59.1%
Correlation vs SPY	0.01	0.65

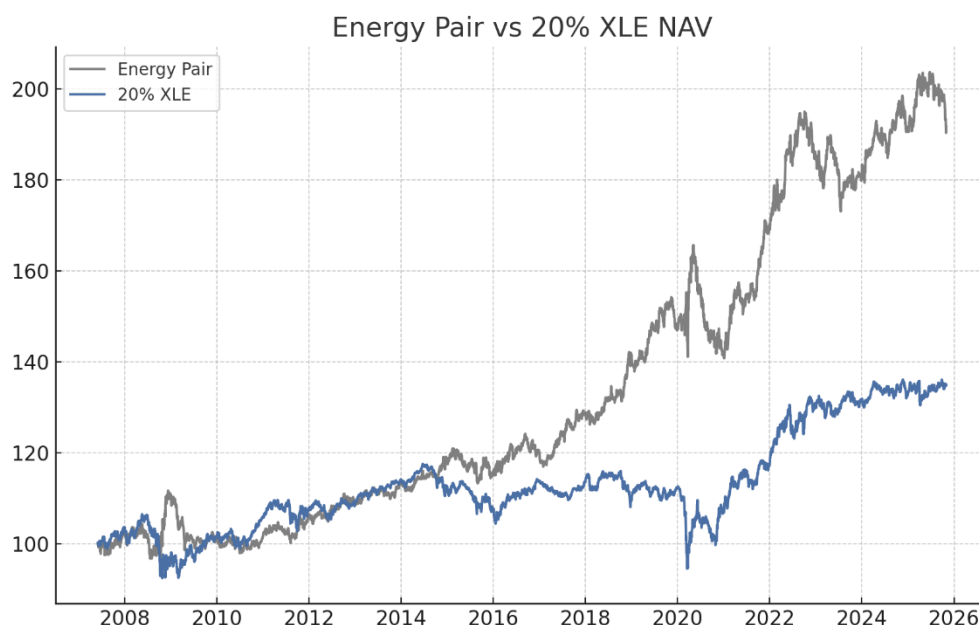


Figure 7: Cumulative NAV demonstrates Energy Pair's conversion of cyclical energy exposure into a genuine portfolio diversifier with equity-like returns.

The NAV comparison shows the Energy Pair's long-term outperformance of 20% XLE position. The Energy Pair (blue) maintains a steadier trajectory, avoiding the long drawdowns that test investor patience and trigger behavioral mistakes. The strategy reaches its destination with a materially smoother ride.



Figure 8: Rolling 12-month returns show Energy Pair's consistent volatility reduction and improved drawdown profile relative to direct XLE exposure.

The rolling 12-month return chart illustrates the volatility compression over investable horizons. XLE (gray) swings wildly from +130% to -60% on a trailing year basis. The Energy Pair (blue) exhibits a dramatically narrower band, rarely exceeding $\pm 25\%$. For

allocators concerned about client experience and behavioral risk, this smoother return profile reduces the likelihood of panic selling at inopportune moments.

Risk Factors

- **Energy Bull Markets:** During sustained energy rallies, the strategy will underperform passive XLE exposure as the short IEZ leg generates losses. The strategy captures only ~6% of upside moves in energy.
- **OFS Outperformance Regime:** If oilfield services companies enter a sustained period of outperformance (major capital cycle upturn, consolidation wave, or extended super cycle), the short IEZ leg will generate losses that may not be offset by XLE gains.

Appendix D: Inter-Pair Correlation & Contribution Analysis

Executive Summary

This analysis examines the correlation structure and return contributions of the three component pair trades within SCI. The findings reveal an asymmetric correlation structure that is central to the strategy risk profile: correlations rise during market stress (when all three pairs activate together) but remain low or negative during SCI drawdowns (when the pairs bleed independently). This pattern differs from traditional portfolio behavior, where diversification fails precisely when it is needed most.

Correlation Analysis

Full-Period Correlations

Daily return correlations across the full sample period (2007–2025) reveal consistently low cross-pair correlations, suggesting diversification benefits.

All cross-pair correlations fall within a narrow band of 0.22 to 0.27, indicating that the pairs respond to largely independent market factors. This low correlation is foundational to SCI risk management properties—when one pair experiences drawdowns, the others are unlikely to suffer proportionally.

	Gold Pair	Energy Pair	Credit Pair
Gold Pair	1.000	0.220	0.266
Energy Pair	0.220	1.000	0.223
Credit Pair	0.266	0.223	1.000

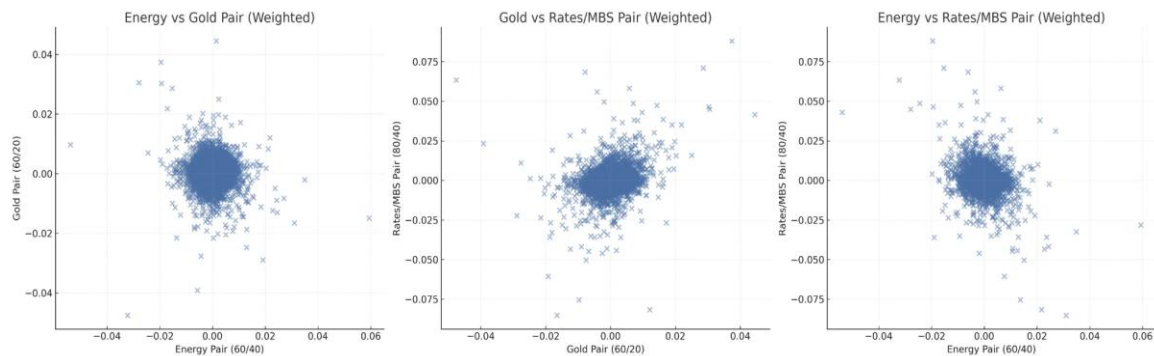


Figure 1: Cross-pair return scatter plots demonstrate weak linear relationships between the three pairs, supporting the diversification benefits of the multi-strategy approach

Crisis–Regime Correlations

Correlation structures shift meaningfully during crisis periods. Importantly, rising correlations during market stress represent a feature rather than a flaw for a convexity portfolio. Unlike traditional long-only portfolios—where correlations approaching 1 during selloffs means diversification fails when needed most—elevated correlations in SCI indicate that all three hedges are activating simultaneously. The pairs are moving up together, not down together.

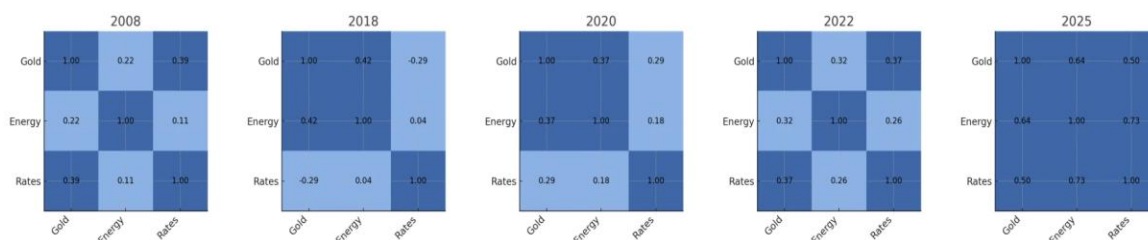


Figure 2: Cross-pair correlations during crisis periods show elevated readings as all three hedges activate simultaneously, representing coordinated protection rather than diversification failure.

- 2008 (Global Financial Crisis): Gold–Rates correlation elevated to 0.39, reflecting simultaneous flight–to–quality and credit spread widening. Energy remained somewhat independent (0.11–0.22 correlations).
- 2018 (Q4 Volatility Spike): Notably, Gold–Rates correlation inverted to –0.29, the only period of negative correlation observed. Energy–Gold correlation spiked to 0.42, suggesting shared exposure to the equity selloff.
- 2020 (COVID Crash): All correlations rose modestly (0.29–0.37 range), as the sudden liquidity crisis created synchronized stress across multiple markets before policy intervention.
- 2022 (Fed Tightening): Correlations remained moderate (0.26–0.37), with the rate–sensitive pairs showing closer association during the historic rate hiking cycle.
- 2025 (Recent Correction): Correlations reached their highest observed levels (0.50–0.73), indicating broad–based hedge activation. All three pairs posted positive returns during this period (Gold +1.3%, Energy +2.7%, Rates +8.6%), confirming that elevated correlation reflected coordinated gains rather than diversification failure.

Crisis Period Contributions

Return attribution by pair during each major stress period demonstrates how different crisis types activate different pairs:

Crisis	Gold	Energy	Credit	SCI Total
GFC (2008)	+21.8%	+4.6%	+52.9%	+79.3%
COVID (2020)	+3.0%	-3.9%	+34.1%	+33.2%
2022 Tightening	+0.6%	+18%	+5.6%	+24.2%
Q4 2018	+1.6%	+4.5%	+5.6%	+11.7%
2025 Correction	+1.3%	+2.7%	+8.6%	+12.6%

Key Observations:

- The Rates/MBS pair dominated during credit-driven crises (GFC, COVID), contributing over 50% and 34% respectively. This reflects the pair's direct exposure to credit spread dynamics and leveraged mortgage arbitrage unwinding.
- The Energy pair provided its largest contribution during the 2022 tightening cycle (+18%), when energy commodity strength combined with capital discipline in the services sector. Notably, Energy was the only pair to post a negative return during COVID (-3.9%), as the oil demand collapse created unusual dynamics.
- The Gold pair contributed meaningfully during the GFC (+21.8%) but has been relatively muted in subsequent crises. This may reflect the pair's higher sensitivity to severe deflationary shocks rather than garden-variety corrections.

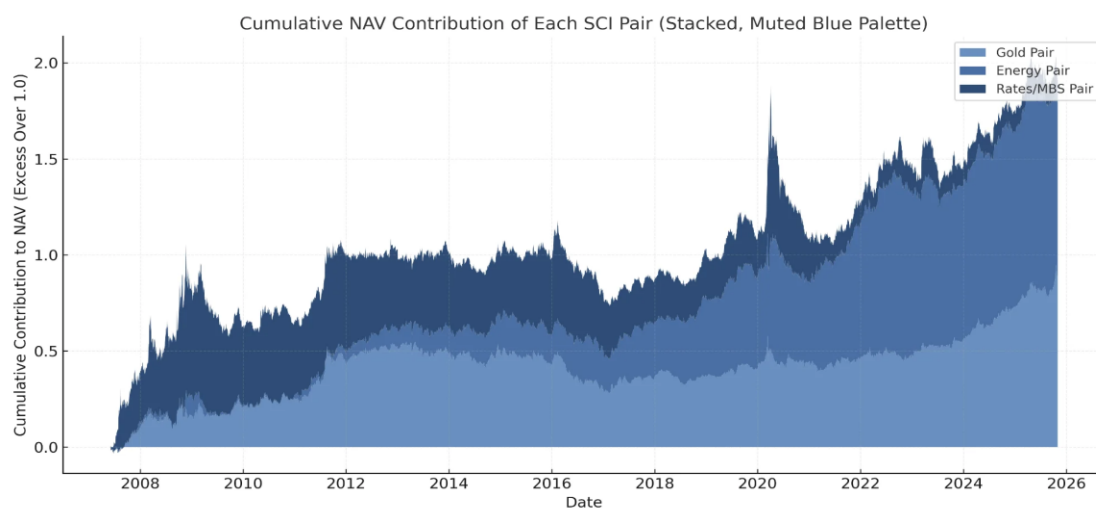


Figure 3: Cumulative NAV contribution by pair demonstrates how different pairs drive returns during different regimes, with the Credit Pair providing the largest crisis-period gains.

Drawdown Characteristics

Individual pair drawdown profiles differ substantially, which is integral to SCI portfolio construction:

- **Gold Pair:** Exhibits the shallowest drawdowns, typically remaining within 10% of highs. This reflects the relatively stable relationship between physical gold and mining equities outside of sector-specific dislocations.
- **Energy Pair:** Moderate drawdown profile with maximum historical drawdowns around 20%. The pair experiences more volatility during energy cycle transitions but recovers relatively quickly.
- **Rates/MBS Pair:** Deepest drawdowns approaching 40% during sustained bull markets in risk assets. This pair pays the premium during good times but delivers the largest payoff during credit stress. SCI maximum drawdown (marked at June 2021) coincides with this pair's deepest underwater period.



Figure 4: Underwater chart reveals staggered drawdown profiles across pairs, with *losses occurring independently rather than simultaneously*.

Asymmetric Correlation Structure

The staggered drawdown profiles are not coincidental—they reflect a structural feature of SCI correlation dynamics. While correlations rise during market stress (when pairs generate gains), they remain low or negative during SCI drawdowns (when pairs are bleeding).

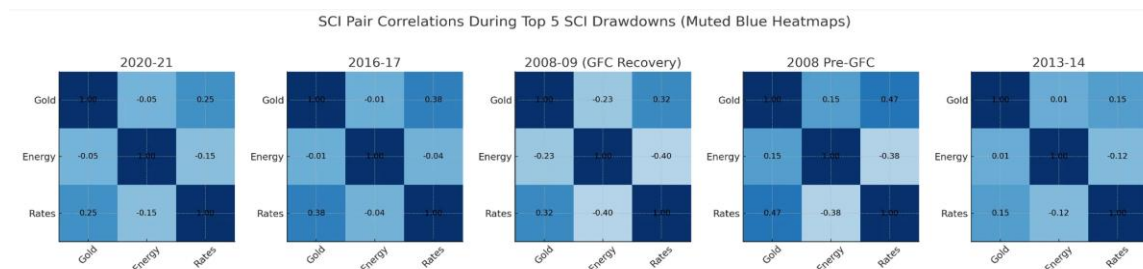


Figure 5: Cross-pair correlations during SCI drawdown periods remain low or negative, confirming pairs bleed independently during unfavorable market conditions.

During the top five SCI drawdown periods, the Energy pair consistently shows negative or near-zero correlation with the other pairs. In the 2008–09 GFC Recovery period, Energy–Rates correlation hit -0.40 . During 2020–21, Energy showed -0.05 correlation with Gold and -0.15 with Rates. This means when one pair is suffering its worst losses, the others are either flat or moving in the opposite direction.

This asymmetric correlation structure—high during gains, low during losses—is precisely the opposite of what occurs in traditional long-only portfolios. It explains why SCI maximum drawdown remains substantially tighter than any individual pair would deliver in isolation, and why the diversification benefits persist exactly when they matter most.

Conclusions

The inter-pair analysis supports SCI multi-strategy construction. Three key findings support the portfolio design:

- **Genuine diversification when it matters:** Cross-pair correlations remain low or negative during SCI drawdown periods, meaning the pairs bleed independently rather than together. This pattern differs from traditional portfolios where diversification fails during stress.
- **Coordinated activation during crises:** Elevated correlations during market stress indicate that all three hedges are working simultaneously—the pairs move up together when protection is needed. This is a feature, not a flaw.
- **Crisis-specific contribution:** Different crisis types activate different pairs most strongly—credit stress triggers Rates/MBS, inflation/supply shocks activate Energy, and severe deflationary events benefit Gold. This means SCI provides protection across a wider range of stress scenarios than any single pair could achieve.

The asymmetric correlation structure—where correlations rise during gains and fall during losses—is central to understanding SCI risk profile. It inverts the traditional portfolio problem of diversification failure during stress, instead providing independent bleed during calm markets and coordinated protection during crises.

Appendix E: Cost and Capacity Analysis

Executive Summary

This Appendix provides an estimate of the costs that would be incurred in implementing SCI. Costs include margin financing, short sale rebates, borrow costs and trading commissions. It also looks at capacity constraints.

Exact costs will vary over time due to fluctuations in benchmark financing rates and borrow rates for short positions. Costs will also vary by prime broker and account size and implementation methodology of short positions (i.e. direct ETF shorting versus shorting underlying holdings).

Total annual costs are estimated to **range from 4.2% at \$1M size for direct implementation via ETFs to 1.64% at \$100M size** for indirect implementation via shorting underlying components. Capacity via direct implementation would be very limited (roughly \$50mm) due to limited availability of shares of REM/MORT. Capacity via indirect implementation would be comfortably over \$2B. These estimates are based on current borrow rates and borrow availability but are very similar to historical average rates and share availability.

It is important to note that these estimates are based on IBKR current and historical rates. Implementation via other brokers will have different costs. However, the principles of costs being lower for larger accounts with indirect implementation should hold true in all cases.

Index Construction

Pair	Long Position	Short Position	Net Exposure
Gold Pair	60% GLD	20% GDXJ	40% Net Long
Credit Pair	80% IEI	40% REM	40% Net Long
Energy Pair	60% XLE	40% IEZ	20% Net Long
Total	200% Long	100% Short	100% Net Long

Gross: 300% (200% Long / 100% Short) • Net: 100% Long

Net Financing Costs

Financing costs consist of margin financing for gross positioning, offset by interest credited on cash from short sales. The table below shows indicative net margin rates by account size for an IBKR Pro account. These are current rates and will fluctuate with benchmark rates.

Account Size	Blended Margin Finance Rate Charged by IBKR	Blended Rate Paid on Short Sale Proceeds by IBKR	Net Margin Rate
\$1,000,000	-4.69%	2.15%	-2.54%
\$5,000,000	-4.45%	3.04%	-1.41%
\$25,000,000	-4.40%	3.32%	-1.08%
\$50,000,000	-4.40%	3.36%	-1.04%
\$100,000,000	-4.27%	3.37%	-0.90%
\$500,000,000	-4.17%	3.39%	-0.78%

Net Margin Rate = Blended Rate Charged + Blended Rate Paid. Negative values represent a net cost. Based on IBKR Pro tiered margin rates as of January 2026.

ETF Implementation

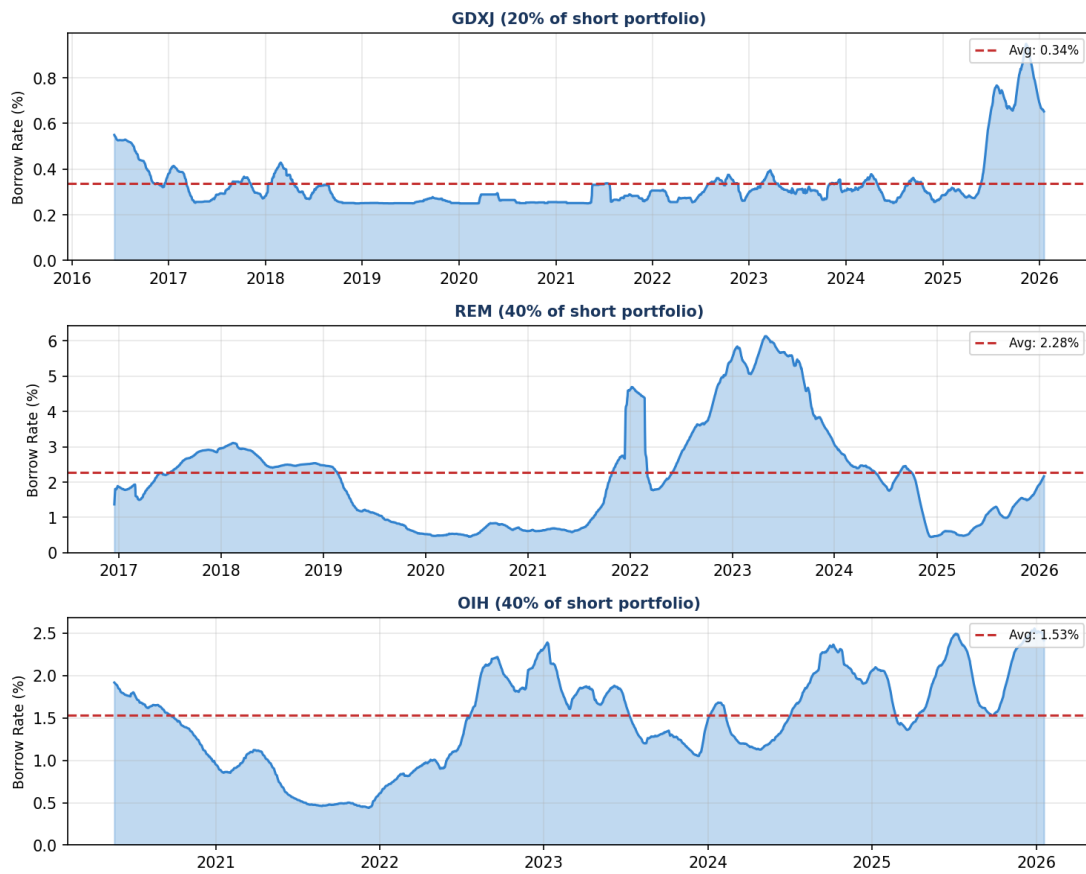
The simplest implementation uses ETFs for all positions. For the long position in gold bullion, using GLDM instead of GLD would save 30 bps per year in expenses. IEI and XLE have very low expense ratios and very high liquidity so there is no reason to look for substitutes. For the Energy short, OIH (VanEck Oil Services ETF) is preferred over IEZ due to greater liquidity and lower borrow costs. Correlation between the two ETFs is 0.99. IEZ and GLD were used for index construction due to their longer price history.

ETF Borrow Rate Statistics

ETF	Short Allocation	Current Borrow Rate	Average Historical Rate	Median Historical Rate	Min Historical Rate	Max Historical Rate
GDXJ	20%	0.58%	0.52%	0.44%	0.25%	2.84%
REM	40%	2.53%	2.28%	2.15%	0.25%	16.34%
OIH	40%	1.14%	1.53%	1.07%	0.25%	13.06%
Weighted	100%	1.68%	1.57%			

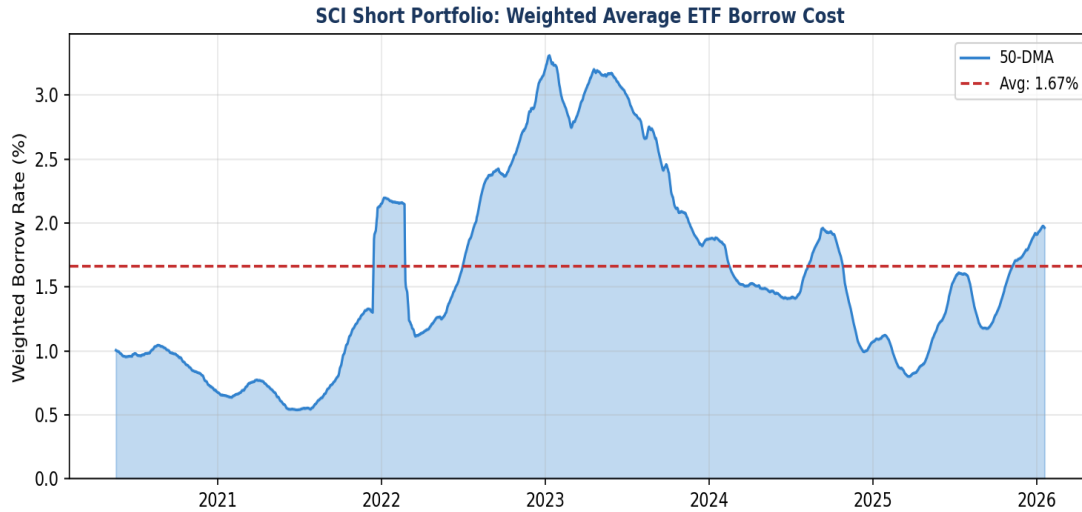
Source: Interactive Brokers borrow rate data, via ChartExchange. July 2015 - January 2026. Current as of January 18, 2026.

Historical ETF Borrow Rates (50-DMA)



Weighted Portfolio Borrow Cost

Current weighted borrow cost: **1.68%** • Historical average: **1.57%**



Component Implementation – Shorting the Underlying

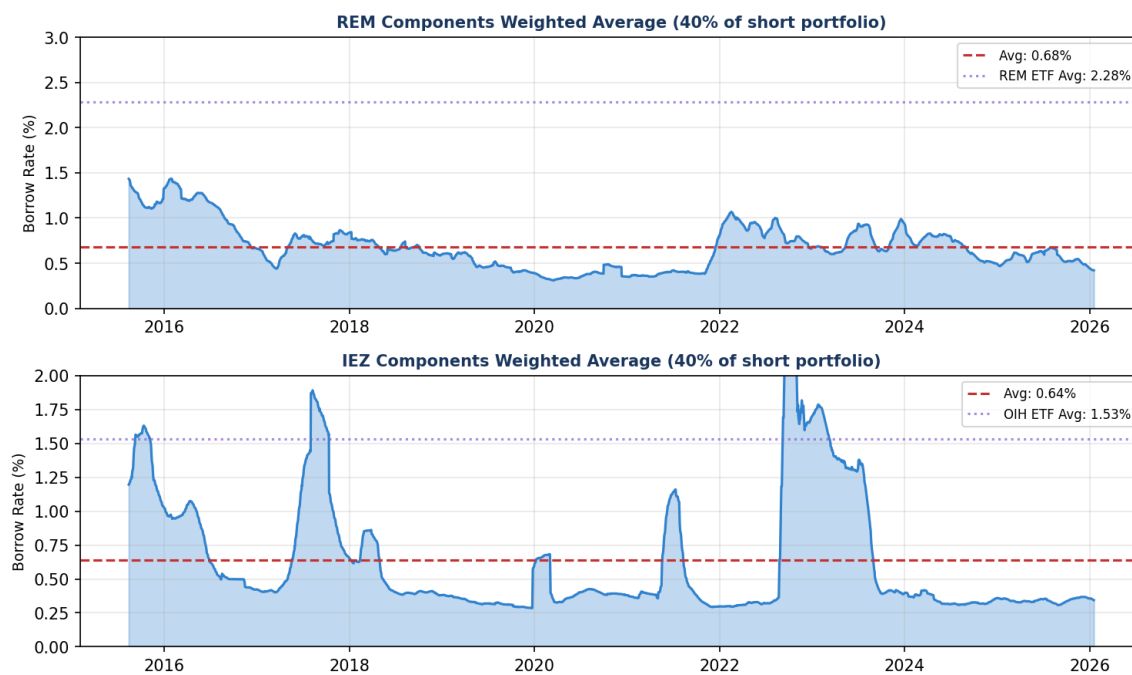
For larger accounts seeking lower costs and greater capacity, an alternative approach would involve shorting the underlying holdings of REM and IEZ/OIH rather than the ETFs themselves. This would require managing approximately 60 individual short positions but would offer cost savings and dramatically increased capacity. The Gold pair continues to use GDXJ ETF in both implementations.

Component Borrow Rate Statistics

Component Group	Allocation	Borrow	Lost ER	Adj. Cost
GDXJ (ETF)	20%	0.58%	n/a	0.58%
REM Components (31 stocks)	40%	0.37%	0.48%	0.85%
IEZ Components (29 stocks)	40%	0.33%	0.38%	0.71%
Weighted Total	100%	0.40%	0.34%	0.74%

Current as of January 18, 2026. Lost ER = foregone benefit from ETF expense ratio decay when shorting components (REM: 0.48%, IEZ: 0.38%). Adj. Cost = Borrow + Lost ER.

Historical Component Weighted Average Borrow Rates (50-DMA)



Purple dotted lines show ETF averages for comparison. Component shorting consistently costs less than ETF shorting.

Borrow Cost Comparison Summary

Current Rates	Gold (20%)	Credit (40%)	Energy (40%)	Weighted Total
ETF Implementation	0.58%	2.53%	1.14%	1.68%
Component Impl. (Adj.)	0.58%	0.85%	0.71%	0.74%
Current Savings	0.00%	1.68%	0.43%	0.84%

Historical Averages	Gold (20%)	Credit (40%)	Energy (40%)	Weighted Total
ETF Implementation	0.52%	2.28%	1.53%	1.57%
Component Impl. (Adj.)	0.52%	1.16%	1.02%	0.98%
Historical Savings	0.00%	1.12%	0.51%	0.65%

Gold pair uses GDXJ ETF in both implementations. Component costs adjusted for lost ETF expense ratio benefit.

Total Implementation Costs

Total implementation cost = Net Margin Cost + Weighted Borrow Cost. The table below uses current finance and borrow rates.

Account Size	Net Margin	ETF Borrow	ETF Total	Comp (Adj.)	Comp Total
\$1M	2.54%	1.68%	4.22%	0.74%	3.28%
\$5M	1.41%	1.68%	3.09%	0.74%	2.15%
\$25M	1.08%	1.68%	2.76%	0.74%	1.82%
\$100M	0.90%	1.68%	2.58%	0.74%	1.64%

Using current financing and borrow rates as of January 18, 2026. Component costs include lost ETF expense ratio adjustment.

Capacity Analysis

ETF-only implementation faces severe capacity constraints, primarily due to limited REM availability. Component shorting dramatically expands capacity by accessing the underlying securities' borrow pools.

ETF Implementation Current Capacity (IBKR Only)

ETF	Shares Avail	\$ Available	Max Fund Size	Status
GDXJ (20%)	3.7M × \$120	\$444M	\$2,220M	

REM (40%)	450K × \$24	\$10.8M	\$27M	BINDING
OIH (40%)	150K × \$314	\$47M	\$118M	

Maximum fund size with ETF implementation: ~\$27M. Maybe \$50M if MORT were also used.

Component Implementation Current Capacity (IBKR Only)

Position	\$ Available	Max Fund Size	Status
GDXJ ETF (20%)	\$444M	\$2,220M	BINDING
REM Components (40%)	\$1,132M	\$2,830M	
IEZ Components (40%)	\$2,894M	\$7,235M	

Maximum fund size with component implementation: ~\$2.2B (44x increase vs ETF)

Trading Commissions

With monthly rebalancing and typical turnover of 10-20% per rebalance, trading commissions would be de minimis relative to financing and borrow costs. At IBKR tiered rates, annual commissions are estimated at less than 0.02% of AUM for ETF implementation and less than 0.05% for component implementation (due to additional tickets). These costs are immaterial relative to net margin costs (~1%) and borrow costs (0.4-1.7%).

Important Disclosures

Data Sources: Borrow rate data from Interactive Brokers, via ChartExchange (July 2015 - January 2026). ETF holdings from iShares and VanEck as of January 2026.

Current vs Historical: Current rates are as of January 18, 2026. Historical borrow rate averages span July 2015 - January 2026. Historical rates assume current constituents for historical period. True historical borrow rates would have to consider actual historical constituents. However, differences are unlikely to be material given that the ETFs are market cap weighted and large holdings have been constituents during the measured period.

Expense Ratio Adjustment: Component implementation costs include an adjustment for the lost benefit of ETF expense ratio decay. When shorting an ETF, the short seller benefits as the ETF loses value to management fees over time. This benefit (0.48% for REM, 0.38% for IEZ) is lost when shorting individual components instead of the ETF.

Financing Estimates: Based on IBKR Pro tiered rates. Actual rates vary by broker. Financing rates will fluctuate over time with changes to the benchmark rate.

Capacity: IBKR availability at a point in time. Institutional prime brokers may have additional inventory.

Index Note: SCI uses IEZ and GLD for backtesting. Implementation may substitute OIH/GLDM for lower implementation costs and/or lower borrowing rates and capacity.